

VT LUNA FALL PRESENTATION

C3: Cosmic Capstone Challenge Track 2

December 4, 2025



Aerospace and Ocean Engineering Dept
Virginia Polytechnic Institute and State University
AOE 4165: Space Vehicle Design
Fall 2025 Semester



Team Organization

Project Lead



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Chief Engineer



Duncan Foster

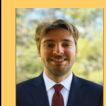
Material



Timmy
Currey



Hugh Young



Ty Brennan



Robert
Higinbotham

Communications, Griffin Lander



Jonah
Bradley



Braeden
Peterson



Eider
Belda Cano



Nicholas
Harvey

Power, Thermal, Energy



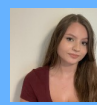
Ty Brennan



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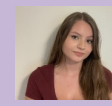


Nicholas
Harvey



Juliana
Evans

Rover Architecture



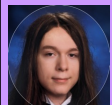
Juliana
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Bradley



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Peterson



Robert
Higinbotham

Stationary System Architecture



Duncan Foster



Timmy
Currey



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Hugh Young



VIRGINIA TECH



Agenda

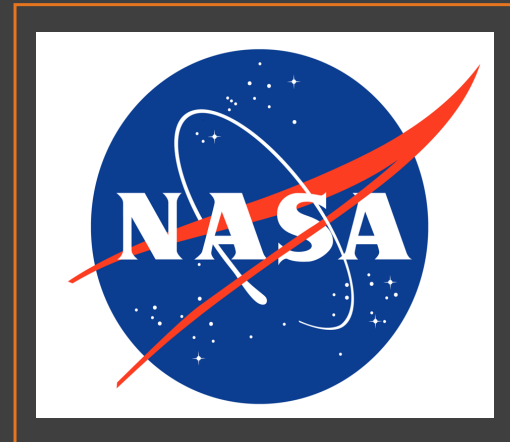
- 1) Project Background and Problem Definition
- 2) Scope and Key Requirements
- 3) System Overview and Concept of Operations
- 4) Systems Synthesis and Analysis
- 5) Architecture Analysis
- 6) Looking Forward



1) Project Background and Problem Definition

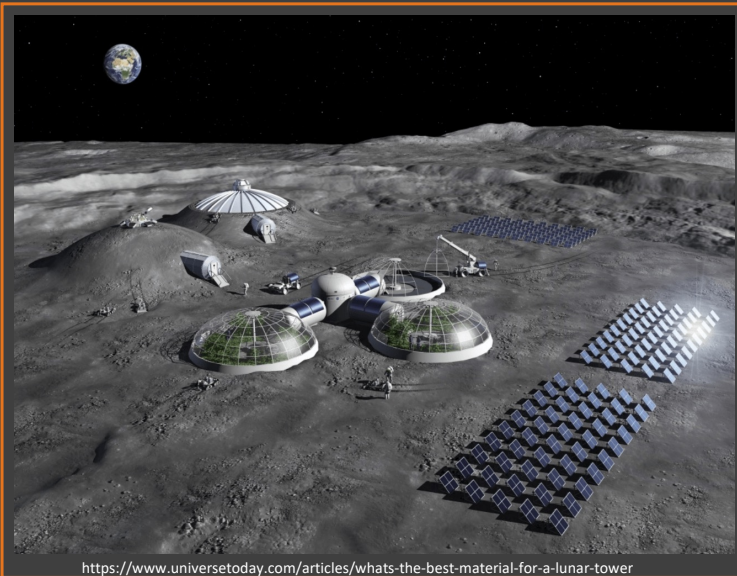
NASA's Artemis Missions

- NASA's Artemis missions were officially approved in 2017 with the intent to bring astronauts back to the moon.
- Through these missions, NASA is focusing on exploring the Moon for technological advancement and to learn how to live on another world while preparing for manned missions to Mars.
- Artemis IV plans on debuting the first lunar space station, Gateway, in 2028.



C3 Cosmic Challenge – Track 2: Lunar Operations

Design a payload for the Griffin lunar lander to help build the infrastructure needed for a permanent lunar outpost



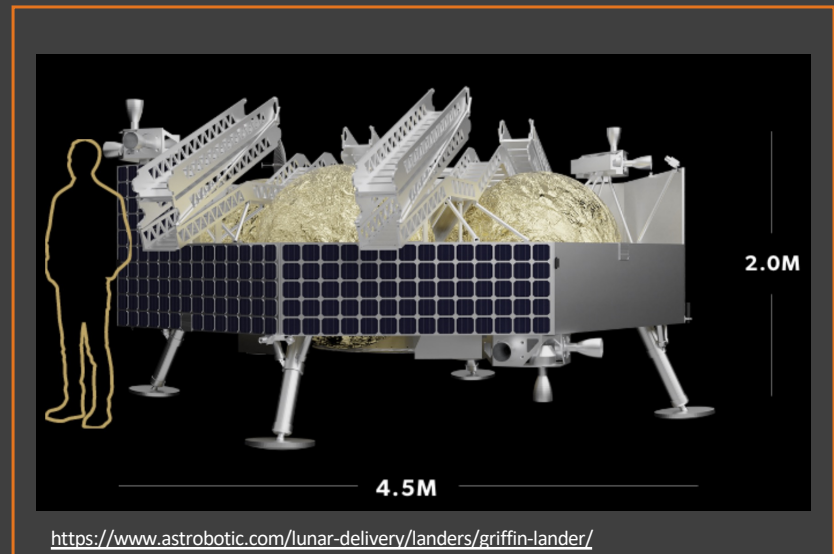
<https://www.universetoday.com/articles/whats-the-best-material-for-a-lunar-tower>



<https://payloadspace.com/lunar-infrastructure-startup-ethos-emerges-from-stealth/>

Astrobotic's Griffin Lunar Lander

- According to the Astrobotic user manual, the Griffin Lunar Lander is a medium-class lander with a maximum payload of 625 kg [Astrobotic].
- The Griffin Lander is very versatile with multiple mounting options for all kinds of payloads [Astrobotic].
- It uses a robust aluminum frame, multiple main engines and attitude/altitude thrusters for descent/orientation [Astrobotic].
- The Griffin Lunar Lander is equipped with five, 3113.76 N, pulsed main engines and twelve, 111.206 N, Altitude Control System (ACS) engines [Astrobotic].
- It makes use of multiple Optical Precision Autonomous Landing (OPAL) sensor systems allowing for it to land in hazardous terrain [Astrobotic].

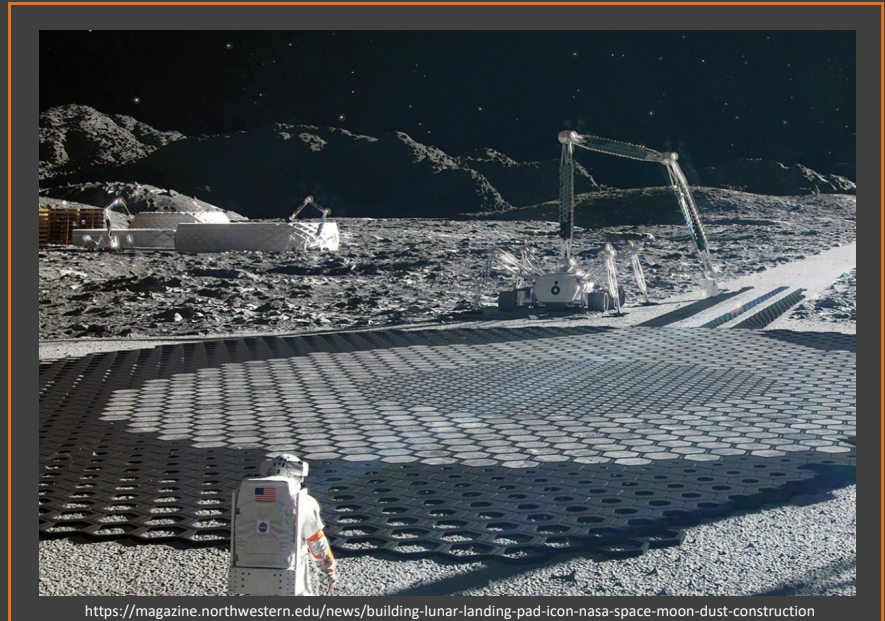


Problem Definition

VT Luna will design a payload that can use lunar regolith to construct lunar landing pads. Using technologies of additive manufacturing and In-Site Resource Utilization (ISRU), this device will create a safe and secure landing pad for future Artemis missions.

Landing Pads

- 'First order' surface infrastructure[NASA].
- Permanent outpost depends on safe, repeatable, onsite landings from large vehicles.
- Lunar dust can be accelerated to over 2 km/s, a velocity high enough to damage hardware even kilometers away[NASA]
- Other forms of infrastructure are all put at risk due to plume ejecta abrasion, and benefit greatly from the advent of on-site material deliveries
- Lunar landing pads mitigate this plume ejecta, stabilize the surface for large vehicle touchdowns, and provide precise reusable landing point.



Timeframe of Mission

Our mission will launch in the late 2020's or very early 2030's to be in line with NASA lunar architecture timeline

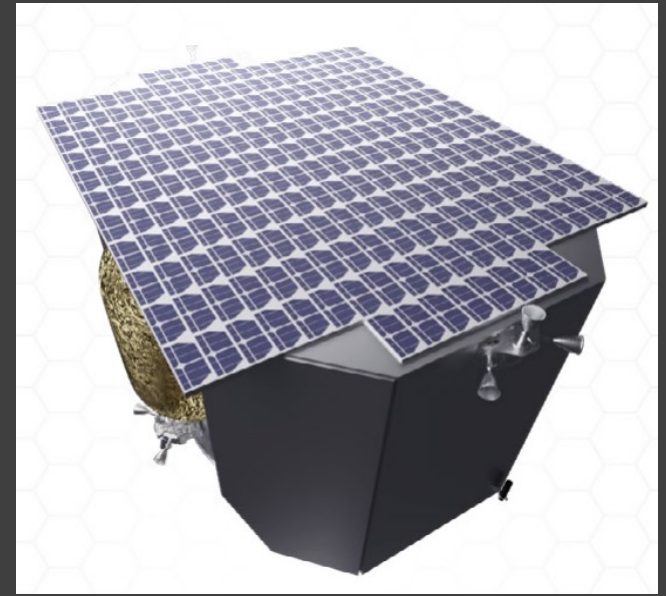
- First preliminary mission to the lunar surface scheduled for 2027, and 2028. Many more missions in the 30's. [NASA]
- Landing pads are inherently one of the first pieces of architecture needed thus will likely be constructed in the early 30's. [Metzger]



<https://spacesettlementprogress.com/tag/lunar-resources/>

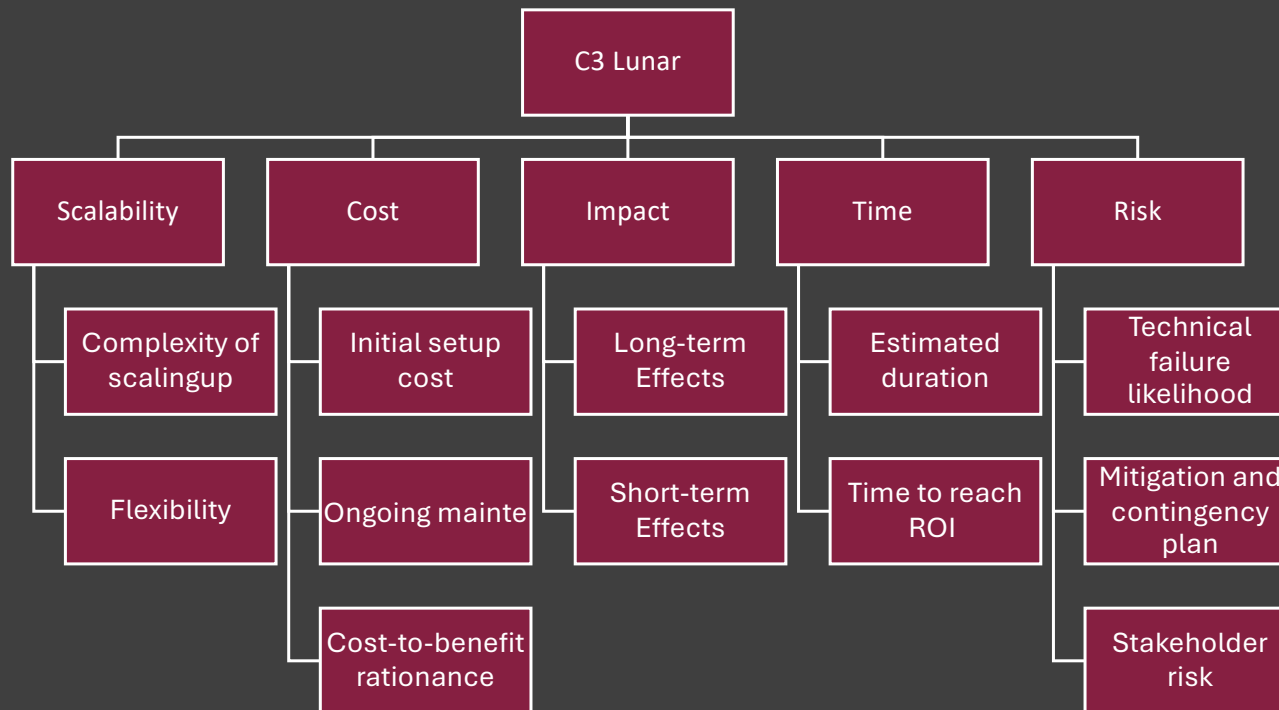
Assumptions

- We are assuming there will be no access to water from the moon (Scaled water extraction architecture unlikely before mid 2030's) [NASA]
- We are assuming we will have 5 kW of power from the Griffin lander [Cosmic]
- We are assuming that there will be a local on-surface astronaut for tile moving / testing [Cosmic]



<https://www.astrobot.com/announcing-lunagrid-a-commercial-power-service-for-the-moon/>

Value System Design



Factors	Weight	Rank
Scalability	0.444	1
Impact	0.292	2
Time	0.129	3
Risk	0.081	4
Cost	0.052	5

2) Scope and Key Requirements

Scope

Goal

- The goal of this project is to demonstrate that launch pad tiles can be fabricated using a rover collection system and ISRU. Once produced, each tile will be evaluated on-site by an astronaut to validate its performance and suitability for operational use.

Future

- In the future, this system can be scaled to construct a full lunar launch pad by increasing production through put, preparing terrain areas, and advancing automation. While the tile-maker is capable of producing sufficient tiles for an entire pad, improvements to the tile placement process will be necessary to achieve a fully autonomous, end-to-end construction workflow

System Requirements

ID	Requirement Description
SYS1.0	The system shall have a total mass of under 200 kg.
SYS2.0	The system shall occupy a volume no greater than 75% (0.75) of the Griffin Lander payload.
SYS3.0	The system shall autonomously manufacture structurally sound tiles using In-Situ Resource Utilization (ISRU).
SYS4.0	The system shall accept and process lunar regolith to use as the primary material for tile production.
SYS5.0	The system shall produce hexagonal tiles that meet defined structural and thermal performance standards.
SYS6.0	The system shall operate autonomously for the duration of the mission, including health monitoring and fault recovery.
SYS7.0	The system shall be capable of surviving and operating within the lunar surface environment (thermal and radiation).
SYS8.0	The system shall perform all operations within the allocated peak power and energy-per-tile budget.
SYS9.0	The system should produce 1 structurally sound tiles autonomously in within 2 weeks.
SYS10.0	The system should detect and respond failures with the Griffin Lander.
SYS11.0	The system should store sufficient regolith to produce a 2-3 tile without additional delivery.

3) System Overview and Concept of Operations

Regolith Processing Methods

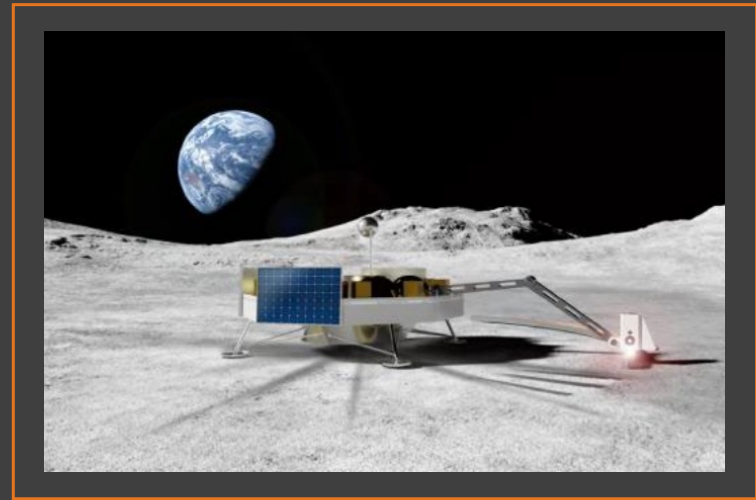
- **Chemical Binders**
 - Require earth supplied liquids
 - Can freeze/boil in a vacuum
- **Solar Sintering**
 - Needs constant direct sunlight
 - Large optics with continuous tracking
- **Heating Plates**
 - Heavy and power hungry
 - Slow heating
- **Pressed Brick**
 - Require high force presses
 - Often uses binders with bad materials like regolith



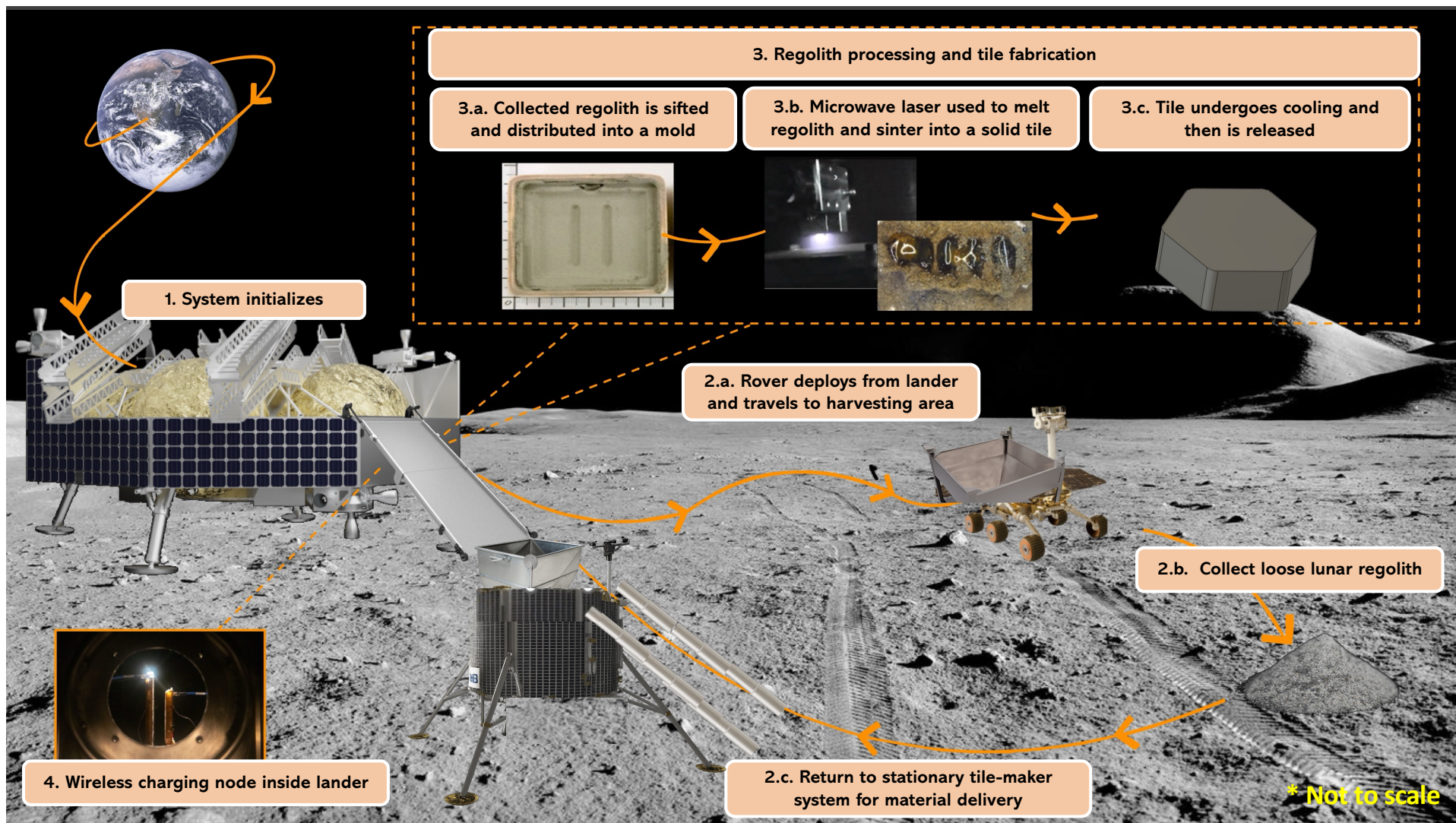
https://1.bp.blogspot.com/-t_hg0II33Wo/X3nn51ObmbI/AAAAAAAAFTM/YZL6ba7QXc8Xva61VRbK9JmjSxbF7mScgCLcBGAsYHQ/s2048/SpaceX%2BStarship%2BBat%2BNASA%2BArtemis%2BBase%2BCamp%2Bspaceport%2Bby%2BICON%2B%2526%2BSEArch_closeup_humanmars.net.jpg

Why Microwave Sintering

- **Reusable**
 - No consumables
 - Same hardware for unlimited tiles
- **Effective**
 - Strong fused regolith pads
 - Creates deep, uniform heating
- **Repairable**
 - Damaged areas are re-meltable
- **Scalable**
 - Infinite growth possibility only limited by power and time
 - Can build lots of infrastructure foundations
 - Landing pads
 - Roads
 - Building foundations
 - Etc.



Preliminary concept of the DM-1 MIMPACT surface construction system operating from the deck of a CLPS lander on the Moon. R.G. Clinton, et.al. 73rd International Astronautical Congress (IAC), Paris, France, 18-22 September 2022.

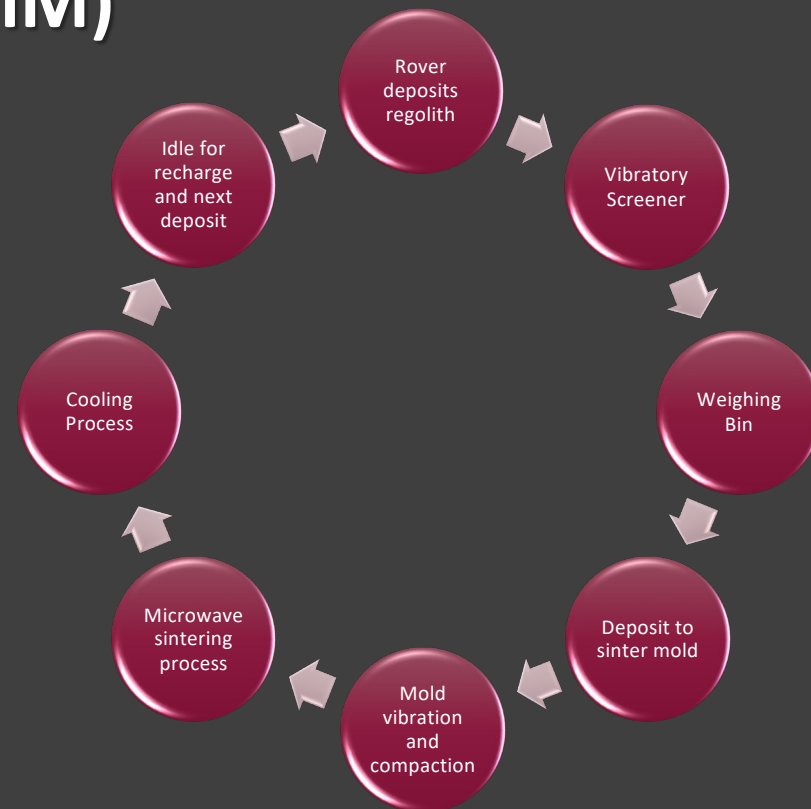


Post Landing and Power Up

- **Landing Configuration**
 - All power systems stored
 - Stationary tile machine and collection rover stowed on lower deck
- **Deployment Sequence**
 - Solar panel unfolds
 - Power and electronics boot and self test
 - Payload is deployed under the 'deployable active' procedure outlined by Astrobotic user manual
 - Deploy comms: 2.4 GHz WLAN modem, compliant with IEEE 802.11n, for local communications. Lander acts as a communications relay from the payload to Earth.
- **Health and Safety Checks**

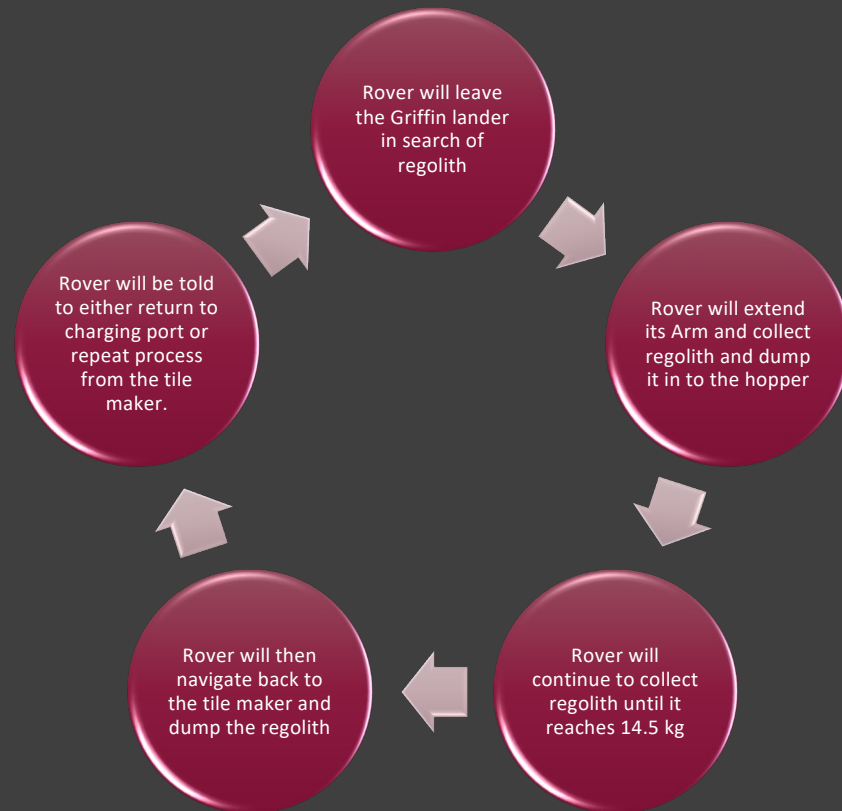
Stationary Tile Maker (STiM)

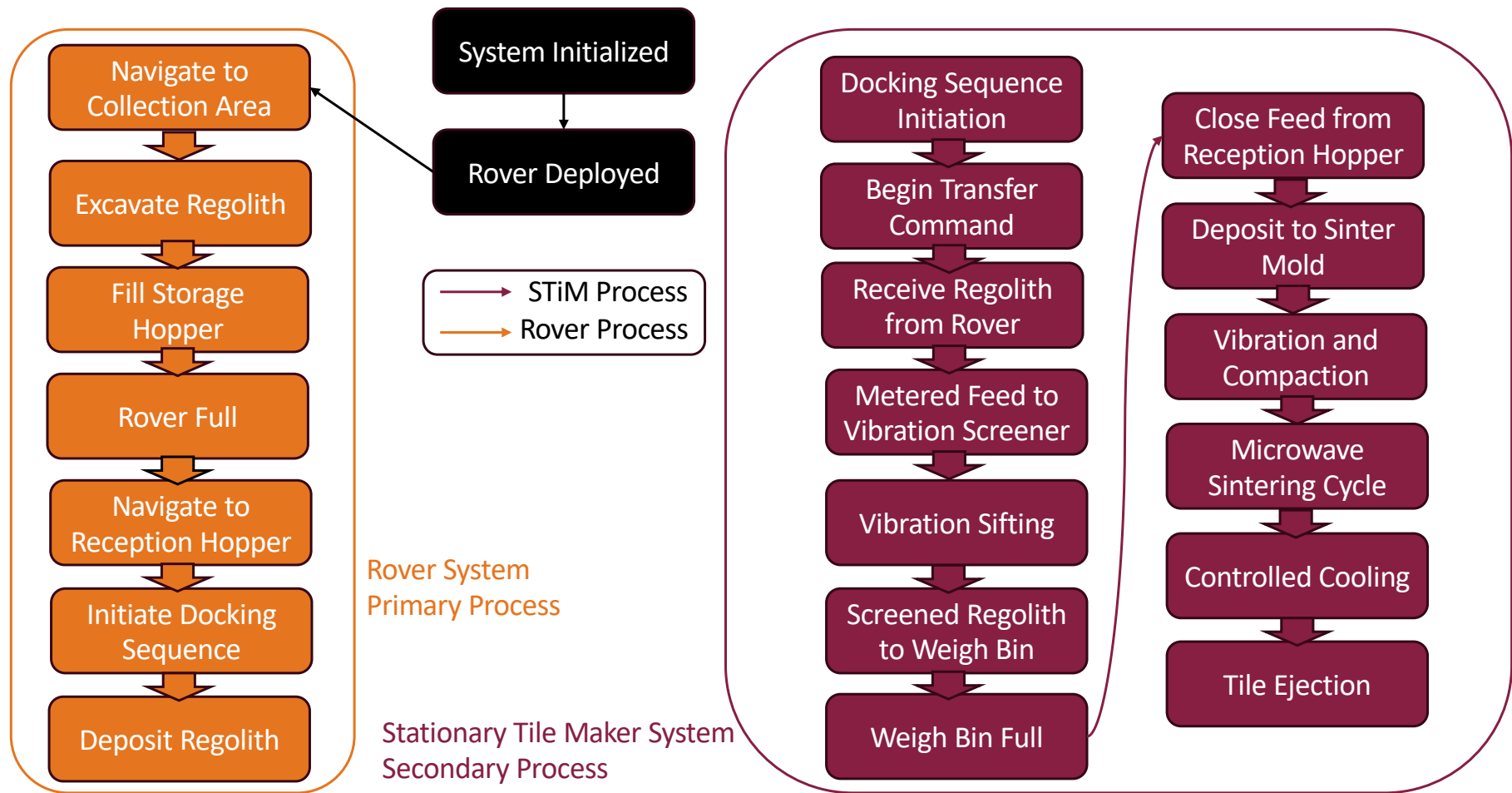
- Stationary Tile Maker will undergo the loop shown to left the right in junction with the rover
- The main purposes of the stationary tile maker are:
 - Storage
 - Conditioning
 - Sintering
 - Cooling

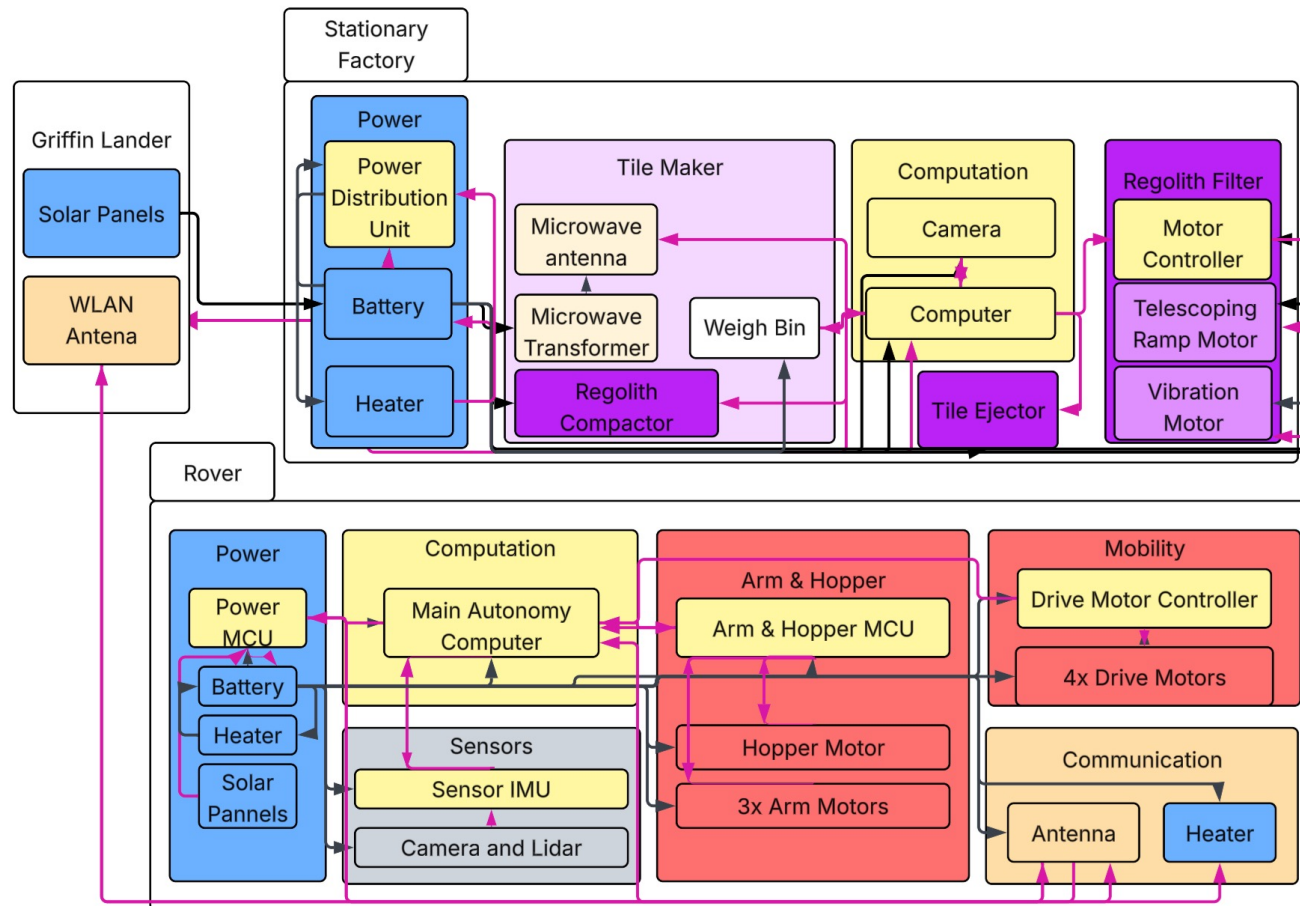


Rover system

- The rover will deploy upon system initialization
- To the right is the process the rover will continually undergo.
- While there are many parts of the rover that are important for this mission, the most are the hopper, the arm, the rover's tires, navigation, and computer systems.







4) System Synthesis and Analysis

Landing Site Selection Criteria

- **Maximum Solar Illumination**
 - Continuous sunlight is essential for power generation
 - South polar regions receive up to 85–90% illumination throughout the lunar year [Bryant, S.H]
- **Safe and Accessible Terrain**
- **Communication Access with Earth**
 - The landing site must maintain line-of-sight to Earth or lunar relay satellites for communication
- **Temperature Stability**
 - Polar highlands experience more stable thermal conditions, reducing thermal cycling and equipment stress
- **Thick and stable composition of the Regolith**
 - The regolith must be suitable for construction techniques such as microwave sintering.

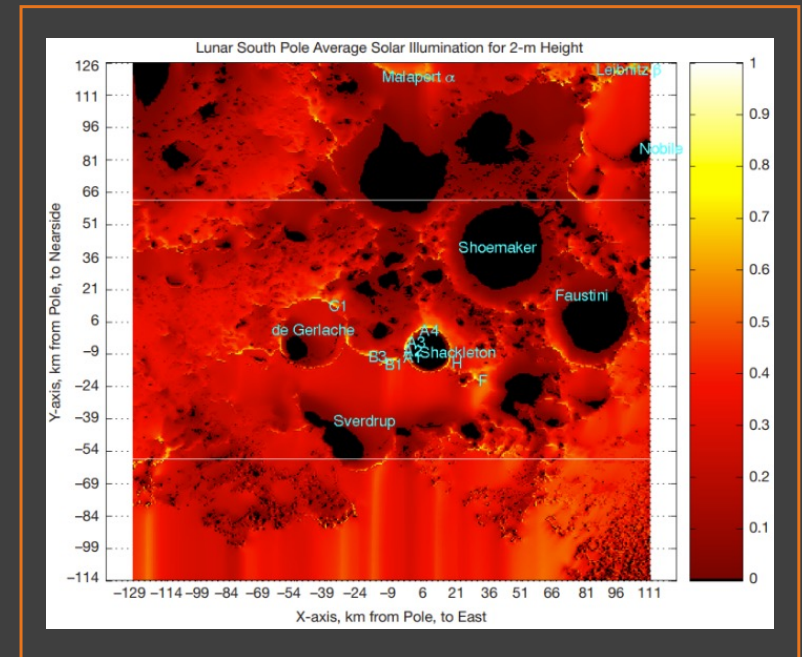


Image courtesy of Bryant, S.H. at "Lunar Pole Illumination and Communications Maps Computed from Goldstone Solar System Radar Elevation Data"

Landing Site Selection

Final Landing Site Selection: **Mons Mouton Plateau**

70-80% annual illumination

Flatter and safer terrain than Shackleton or De Gerlache rims

Reliable line-of-sight communication with Earth

Close to PSRs (Haworth, Nobile, de Gerlache) for water-ice access

Thick, stable and predictable regolith composition

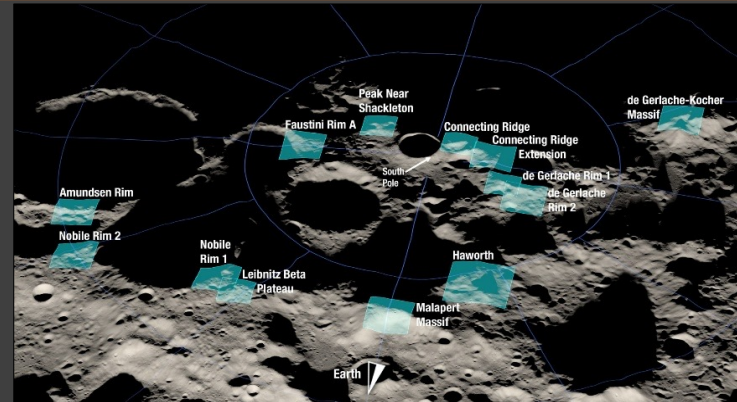


Image courtesy of NASA (<https://www.nasa.gov/news-release/nasa-identifies-candidate-regions-for-landing-next-americans-on-moon/?utm>)

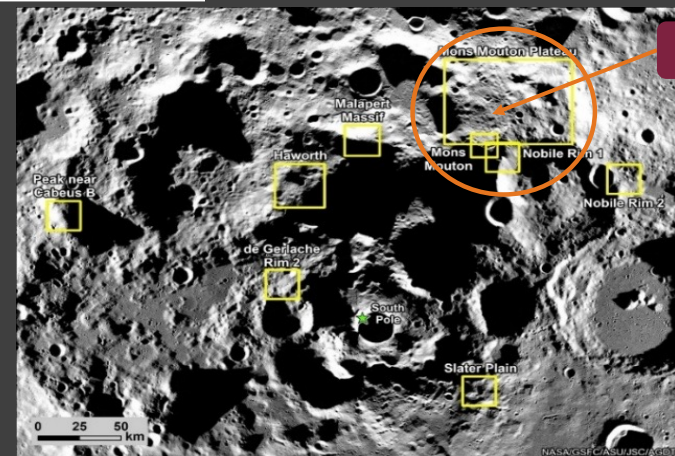
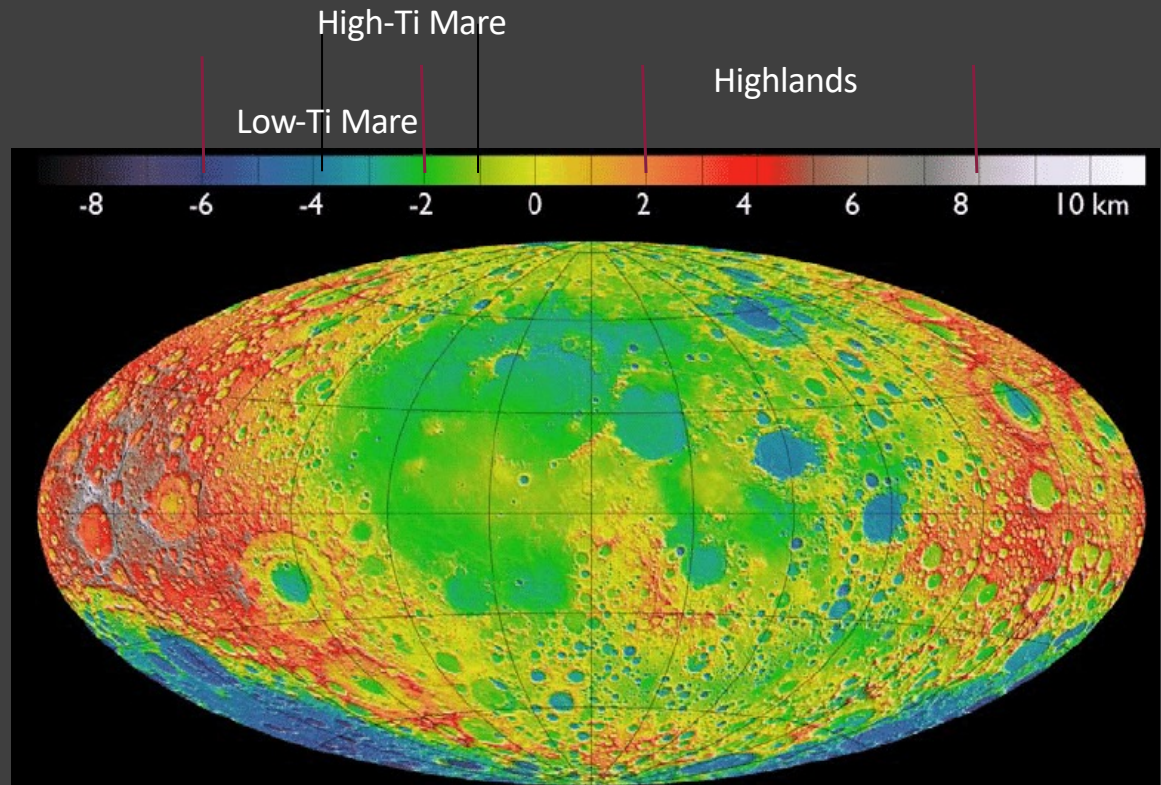


Image courtesy of NASA (<https://www.nasa.gov/news-release/nasa-provides-update-on-artemis-iii-moon-landing-regions/#?text=Peak%20near%20Cabeus%20B,with%20Earth%2C%20and%20lighting%20conditions>)

Nobile Rim 1

Lunar Regolith Mare Dispersion

- 3 main sample types analyzed:
 - Lunar Highland 2 – 8 Km
 - High-Ti Mare -4 - -1 Km
 - Low-Ti Mare -6 - -2 Km



https://www.researchgate.net/figure/Lunar-topographic-map-with-the-first-1-billion-LOLA-measurements_fig2_224320991

Lunar Regolith Properties – Chemical Make Up

Chemical Compound	Concentration (%)
Silicon Dioxide	48-42
Titanium Dioxide	7-1
Aluminum Oxide	27-12
Iron Oxide	18-4
Magnesium Oxide	11-4
Calcium Oxide	17-10
Sodium Oxide	0.7-0.4
Potassium Oxide	0.6-0.1
Manganous Oxide	0.2-0.1
Chromium Oxide	0.4-0.2

Lunar Regolith Mare Chemical Make Up

	Highlands (%)	High-Ti (%)	Low-Ti (%)
Silicon Dioxide	44.9	41.0	41.1
Titanium Dioxide	0.5	8.5	2.9
Aluminum Oxide	25.1	12.4	13.1
Iron Oxide	6.2	16.6	17.0
Magnesium Oxide	7.5	8.9	9.9
Calcium Oxide	14.9	11.4	11.7
Other	0.9	1.2	0.3

Landing Site Regolith Properties

High-Ti Mare in the Mons Mouton Plateau

High in Silicon Dioxide as well as Titanium Dioxide and Iron Oxide

	High-Ti (%)
Silicon Dioxide	41.0
Titanium Dioxide	8.5
Aluminum Oxide	12.4
Iron Oxide	16.6
Magnesium Oxide	8.9
Calcium Oxide	11.4
Other	1.2

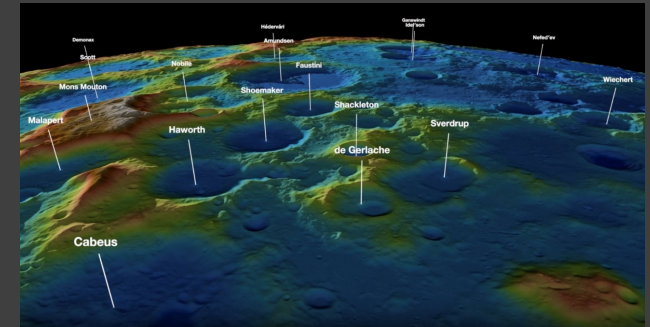


Image courtesy of NASA (<https://svs.gsfc.nasa.gov/5127>)

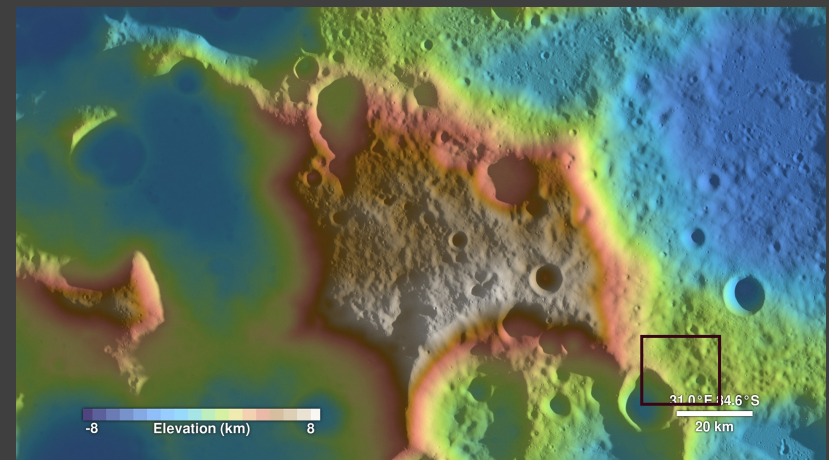


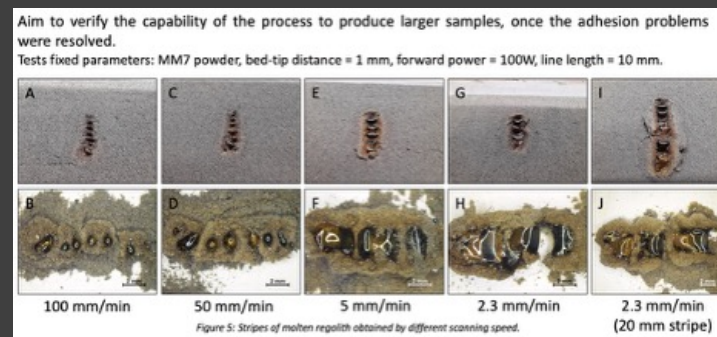
Image courtesy of NASA (<https://svs.gsfc.nasa.gov/5074>)

Chemical Compound Temperature Properties

Chemical Compound	Boiling Point @ 1 atm (K)	Vacuum Sublimation Temp (K)
Silicon Dioxide	3223	2073
Titanium Dioxide	3245	2173
Aluminum Oxide	3250	2073
Iron Oxide	3273	1900
Magnesium Oxide	3873	2173
Calcium Oxide	3123	1723
Sodium Oxide	2223	973
Potassium Oxide	1273	572
Manganous Oxide	3073	1773
Chromium Oxide	4273	2273

Regolith Particle Size and Density

- Using mock regolith particle size and density were analyzed
- Sintering with average grain size $< 600 \mu m$ yield the best results



Master material	MM1		MM2		MM3		MM4		MM5		MM6		MM7		MM8	
Specimen code	340	342	345	346	373	374	351	353	356	358	366	367	376	378	368	370
Measured density (g/cm ³)	1.115	1.060	1.120	1.189	1.273	1.280	1.357	1.334	1.368	1.164	0.970	1.387	1.377	1.382	1.762	1.714
Average density for Master Material (g/cm ³)	1.088±0.027		1.155±0.035		1.277±0.003		1.346±0.011		1.266±0.102		1.179±0.209		1.379±0.003		1.738±0.024	

Table 1 – MM Samples Densities

Regolith Properties – Sintering Products

- Fused silicon dioxide at 1200 °C
- Hydrogen and helium deposited by solar wind will sublime off (90%) at 200-700 C
- 10.1% mass loss after sintering at 1200 °C
- In vacuum boiling points are lowered
- Will boil off gas leading to a porous tile weaking if not accounted for
- Iron will boil off at 1150 K (876.85 °C) in a vacuum – if heated for over 96 hours, 50% of mass will be lost

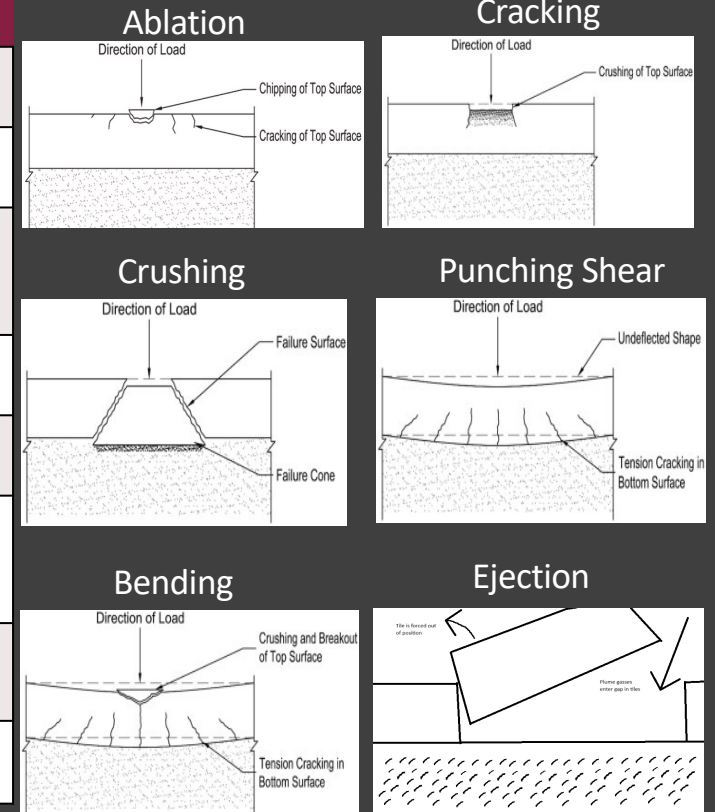
Regolith Properties – What we are looking for

- SiO₂ concentration
- High TiO₂ and FeO concentrations for initial heating
- Low in other contaminants
- Particle size averaging $< 600 \mu m$

Regolith Tile Properties - Requirements

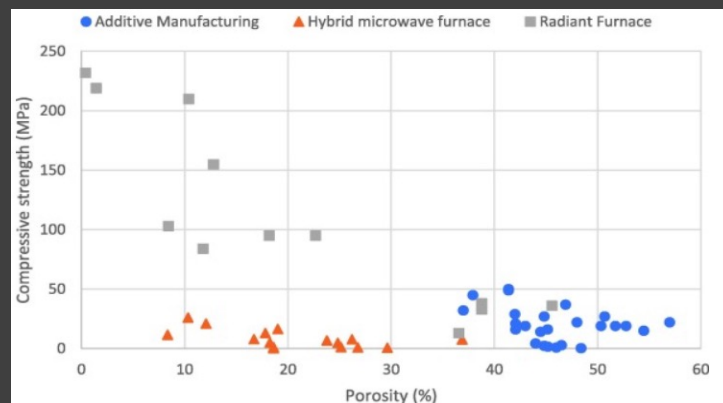
Modes of Failure

ID	Requirement Description
RTP1.0	The tiles shall be homogenous vertically due to manufacturing process
RTP2.0	The tiles shall be a geometry that tessellates
RTP3.0	The tiles shall be able to withstand a sustained vertical pressure of 35.4 kPa to survive the plume pressure of a Starship Human Landing System (HLS) landing
RTP4.0	The tiles shall withstand 423.99 kPa impact of HLS landing*
RTP5.0	The tiles shall withstand 25.78 kPa static pressure of HLS*
RTP6.0	The tiles shall withstand 17786.3 kW/m ² heating from plume of HLS landing*
RTP7.0	The system shall be capable of surviving and operating within the lunar surface environment (thermal and radiation).
RTP8.0	The system shall perform all operations within the allocated peak power and energy-per-tile budget.

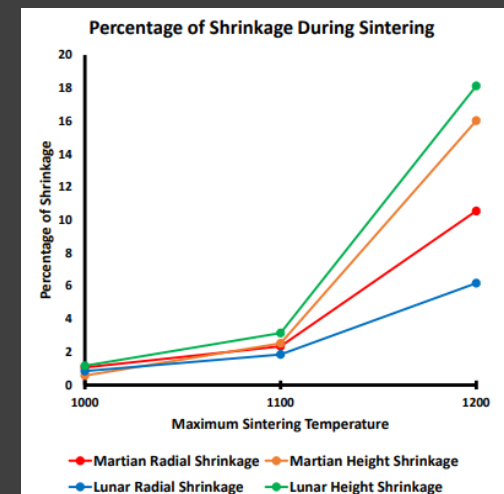


Regolith Tile Properties – Sintering Effects

- Tiles will be sintered at 1200 °C
- The porosity of the tiles will greatly affect the strength
 - Ensuring that only very fine grains are used to make the tiles will decrease the porosity of the final product
- The tiles will go through significant shrinkage during sintering
 - The tile will go through height shrinkages as high as 20%
 - Expected radial shrinkages of approximately 6%



[Lunar Rover Arm Brings Increased Functionality to space missions.](#)



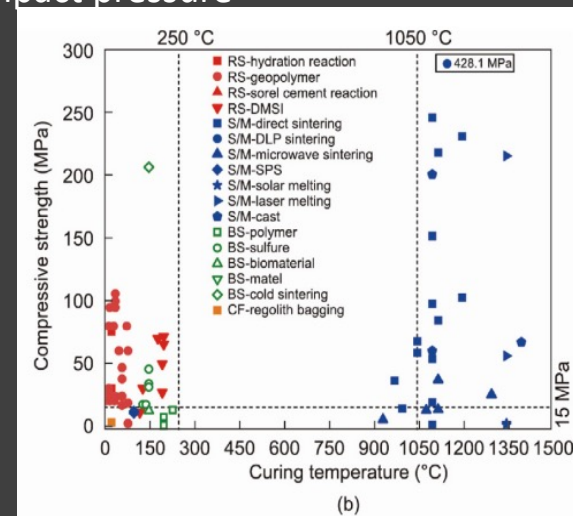
[Lunar Rover Arm Brings Increased Functionality to space missions.](#)

Regolith Tile Properties – Strength

- Tiles need to sustain impact pressure of 424 kPa from the HLS landing leg
 - Differences in the composition of samples and simulants cause the tested properties from different sources to vary
 - The measured compressive strength ranged anywhere from 20 to well over 100 MPa
 - At the lower end of measured compressive strength for lunar samples (20 MPa) a 0.3 m wide (Minimum width of our hexagonal design) tile would have to be 0.106 m thick to have its static compressive strength reach 1.5 times the maximum impact pressure

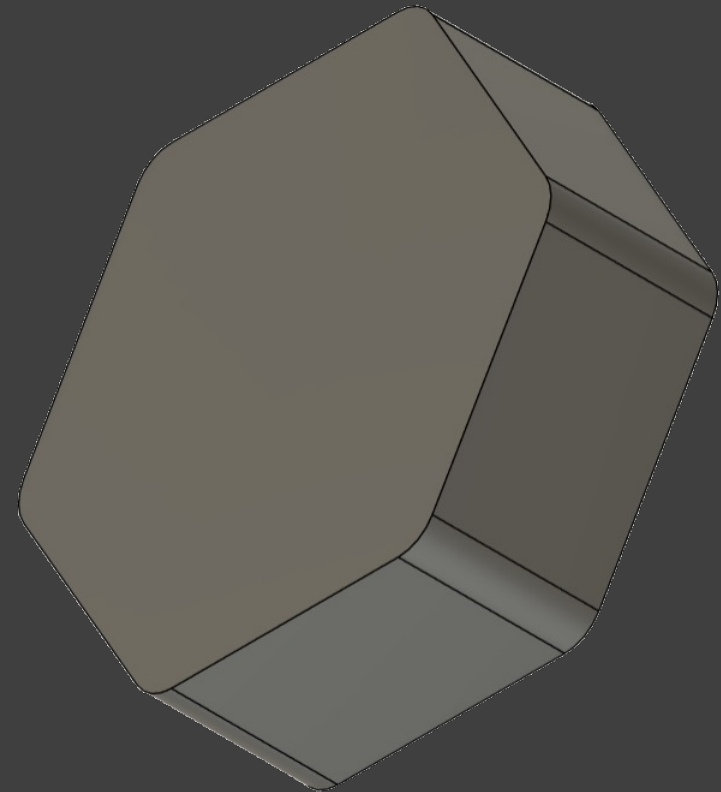
Table 1: Material properties of sintered Lunar and Martian samples.

	Martian Sample	Lunar Sample
1000°C Radial Shrinkage Percentage	1.06%	0.84%
1000°C Height Shrinkage Percentage	0.57%	1.16%
1100°C Radial Shrinkage Percentage	2.34%	1.85%
1100°C Height Shrinkage Percentage	2.52%	3.15%
1200°C Radial Shrinkage Percentage	10.54%	6.17%
1200°C Height Shrinkage Percentage	16.03%	18.13%
1200°C Compression Modulus (MPa)	67.80	55.73
1200°C Compressive Strength (MPa)	25.46	21.73
1200°C Compression Modulus (Ksi)	9.83	8.08
1200°C Compressive Strength (Ksi)	3.70	3.15



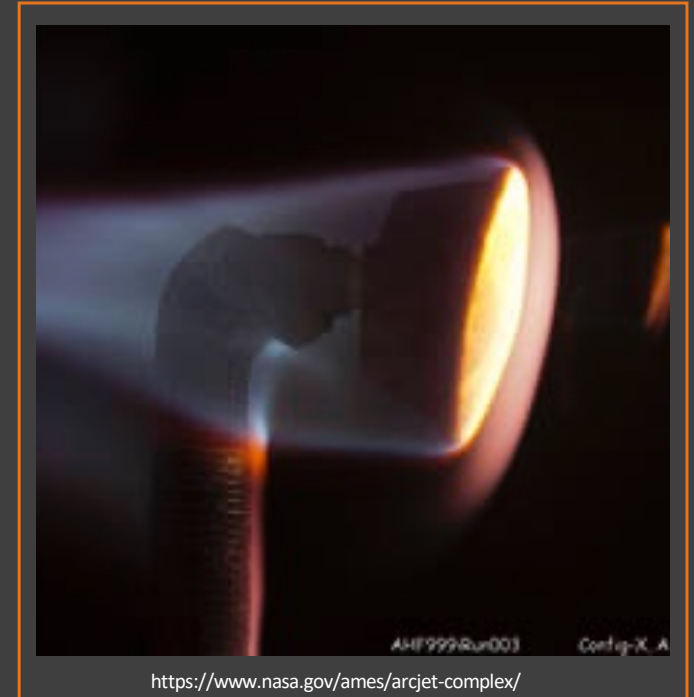
Regolith Tile Design

- A normal hexagonal tile with a circumscribed radius of 0.150 m, a depth of 0.106 m, and a fillet of 25 mm was chosen
- The radius is based upon the necessary surface area (0.078 m^2) to withstand the load of a landing while being small enough to distribute landing pad load to multiple tiles*
- Quadrilaterals and triangles were avoided as sharp angles are too prone to stress concentration
- Polygons with more sides were avoided as they do not tessellate



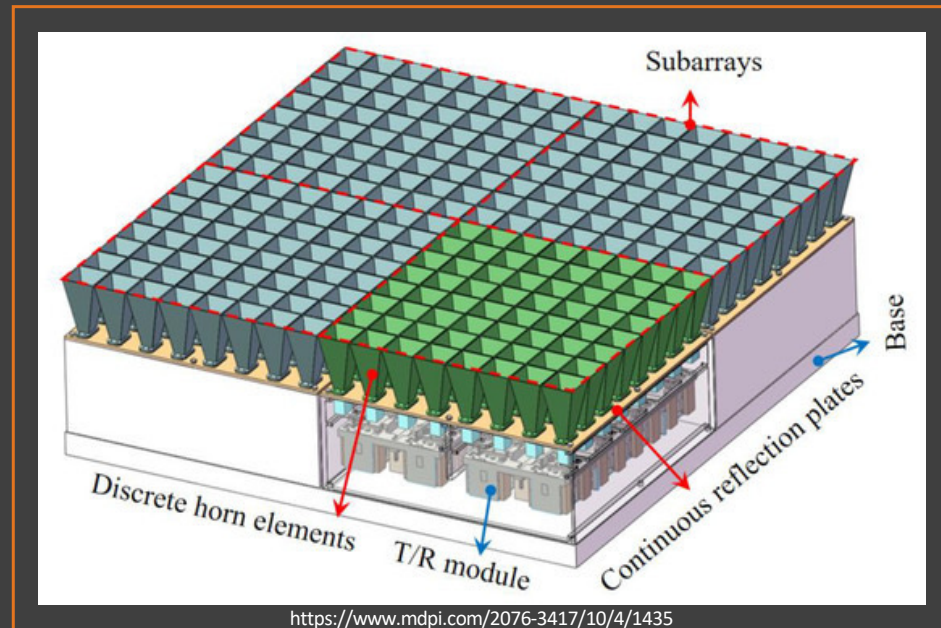
Testing of Tile

- We will use an astronaut to retrieve the produced tile and return it to Earth to undergo testing.
- A hydraulic press will be used to measure the maximum force and in turn pressure the tile can withstand. [SSHL Machinery]
- A hydraulic actuator will be used to test the impact pressure the tile can withstand. [Maverick]
- An Arc jet will be used to mimic the plume of a rocket and test the thermal capabilities of the landing pad. [NASA]



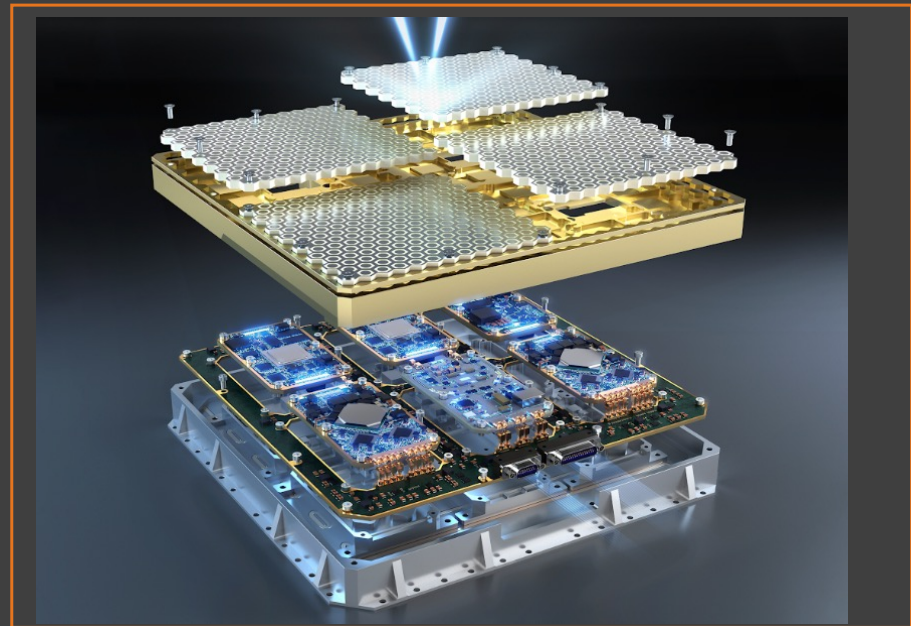
Microwave Sintering Generator and Applicator

- 1 kW 915 MHz sintering head/plate
 - 915 MHz has better penetration compared to 2.45 GHz
 - 2.45 GHz shows better power efficiency
 - 915 MHz won't interfere with WLAN
- 1 kW microwave generator like the Wattsine WSPS-915MHz-1kW
- Hexagonal Phase array antenna
 - Allows for splotch sintering
 - Can systematically sinter each layer



Phased Array Sintering

- Full sized phased array brick
 - Encompasses the entire hexagon
- Electronic beam steering
 - We move the hotspot using the array
 - No mechanical complexity or fail points
 - Maintains a high E field
 - Uniform Heating



<https://payloadspace.com/steering-the-future-with-active-phased-arrays/>

Sintering Power Model - Variables

Variable	Symbol	Unit	type	Dependencies
Power Density	P	W/m^3	Found	K,f,E,k',tan(δ)
Power draw	P_s	W	Found	P, η , V
Energy Required	J	kWh	Found	P_s , t
Heating rate	ΔT	$^{\circ}C/s$	Found	P, ρ , CP
Heating time	t	s	Found	T_0 , T_t , ΔT
Instant Temperature	T	$^{\circ}C$	Found	T_0 , ΔT
Initial Temperature	T_0	$^{\circ}C$	Defined	N/A
Target Temperature	T_t	$^{\circ}C$	Defined	N/A
Dielectric Constant	K'	N/A	Material Property	T, f
Loss Tangent	Tan(δ)	N/A	Material Property	T, f
Density	ρ	kg/m^3	Material Property	T
Specific Heat Capacity	CP	J/kgK	Material Property	T

Variable table part 1 (Found values, defined values, and material properties)

Variable	Symbol	Unit	type	Dependencies
Tile Area	A_t	m^2	Tile Property	N/A
Layer Thickness	h_l	m	Tile Property	N/A
Layer Count	n_l	N/A	Tile Property	h
Tile Thickness	h	m	Tile Property	h_l , n_l
Spot Count	n_s	N/A	Tile Property	A
Tile Volume	V	m^3	Tile Property	A, h
Applicator Area	A_a	m^2	Hardware Property	N/A
Microwave frequency	f	Hz	Hardware Property	N/A
Electric Field Strength	E	V/m	Hardware Property	N/A
System Efficiency	η	N/A	Hardware Property	N/A
Free Space Permittivity	ϵ_0	F/m	Constant	N/A
Equation Constant	K	F/m	Constant	ϵ_0

Variable table part 2 (tile properties, hardware properties, and constants)

Sintering Power Model - Equations

- Needed in order to obtain power consumption, hardware properties, and construction timeline
- Used for battery and additional solar panel size as well as energy budgeting

- Primary Equations [NASA]:

- $P = K f E^2 k' \tan(\delta)$

- $\Delta T = \frac{P}{\rho C P}$

- $P_s = \frac{PV}{\eta}$

- $K = 2 \cdot \pi \cdot \epsilon_0$

- $V = A_t \cdot h = h_l \cdot n_l \cdot A_a \cdot n_s$

- $t = \frac{T_t - T_0}{\Delta T}$

- $J = P_s \cdot t$

- Cole-Cole Model [NASA]:

- $K = K' - jK'' = K_\infty + \frac{K_0 - K_\infty}{1 + (j\omega\tau)^{1-a}}$

(Only equation not using reference variables)

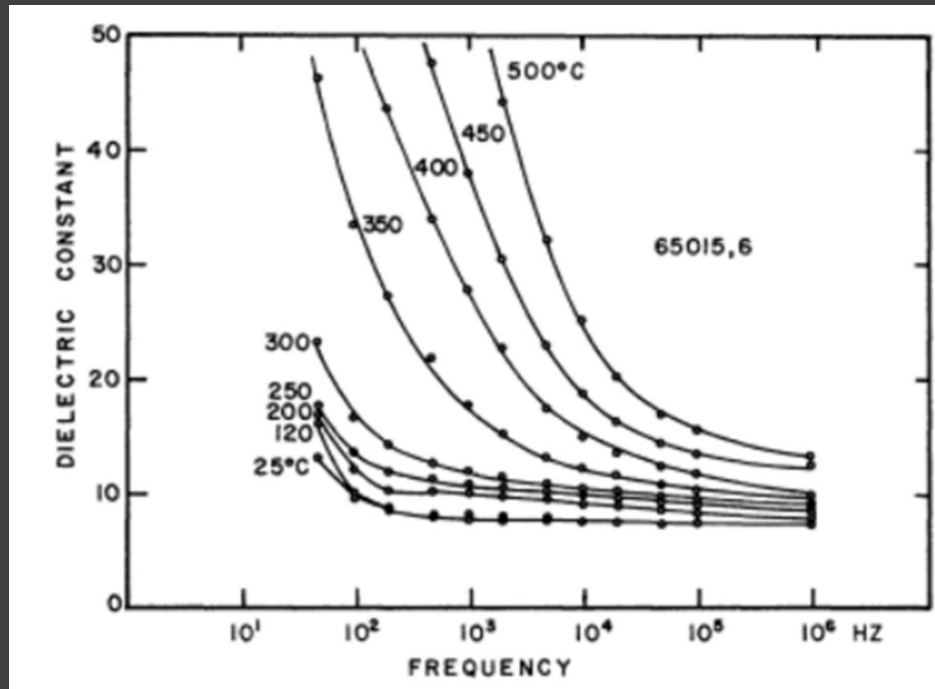
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<https://articles.adsabs.harvard.edu//full/1973LPSC....4.3133O/0003138.000.html>

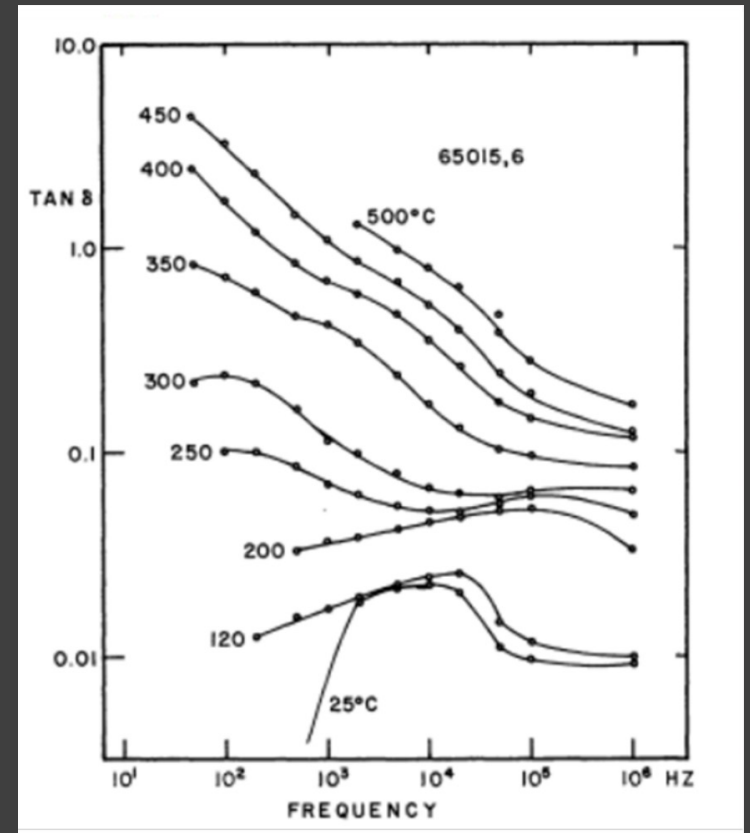
Sintering Power Model - Method

- Conservative estimates
 - 25% system efficiency
 - 475 J/kgK Specific Heat Capacity [Kost, Linke]
 - 2000 kg/m^3 density [Bassett, Shackelford]
 - 1200 C° Target Temperature [Kim, Ryu]
- Iterative while loop for differential equations
- Heat dissipation
- Applicator area sweep
- Point extraction for multivariable curve fitting

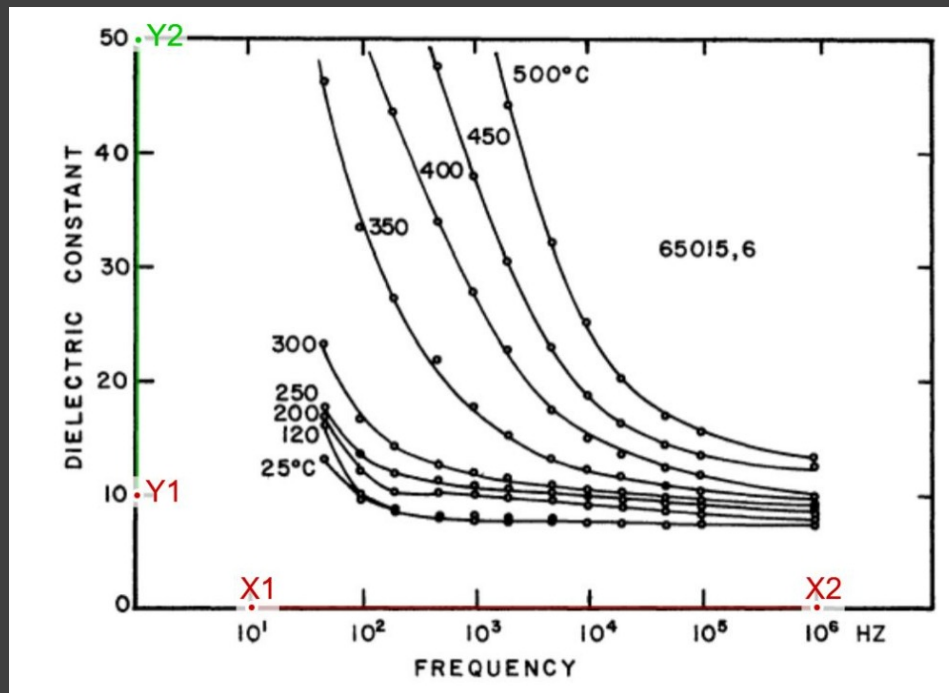
Sintering Power Model - Method



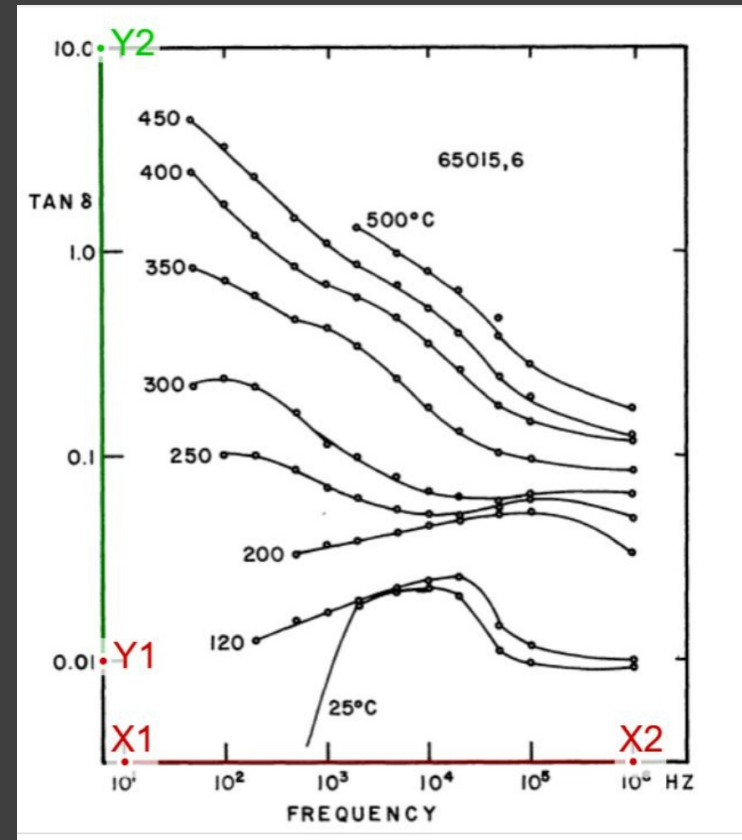
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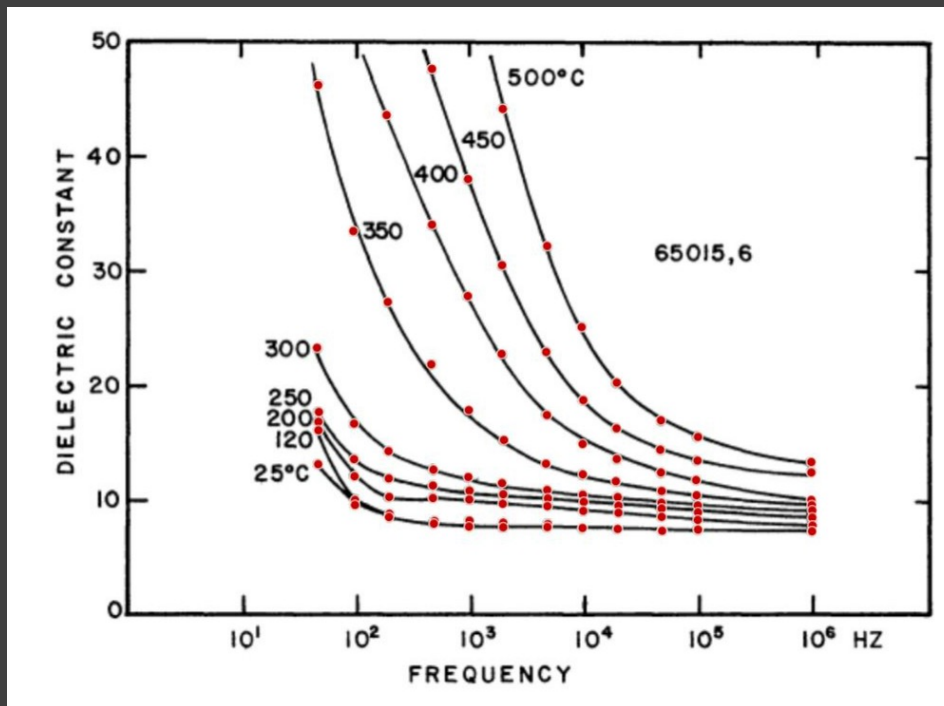
Sintering Power Model - Method



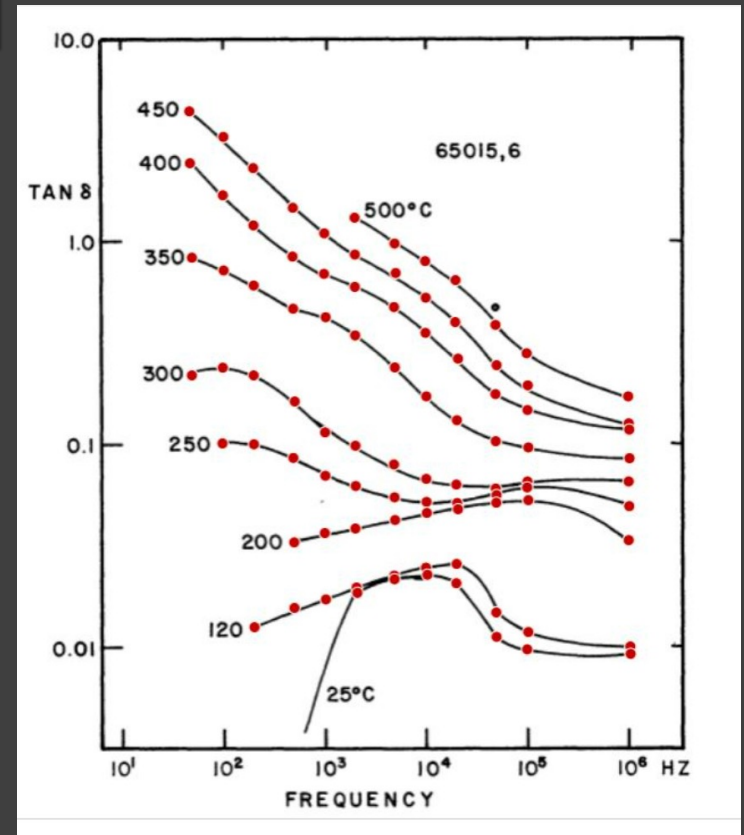
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Sintering Power Model - Method

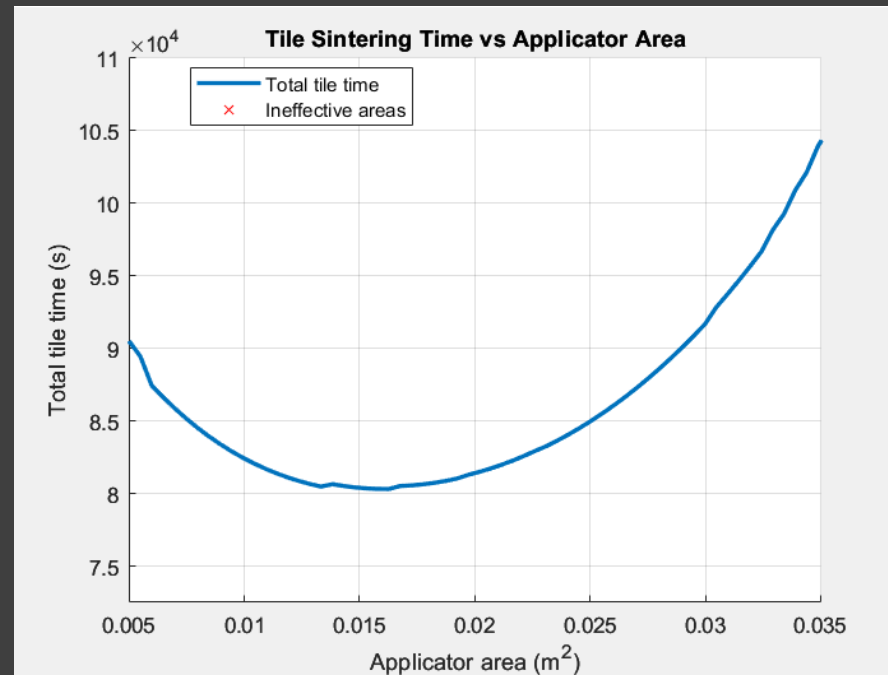


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Sintering Power Model - Findings

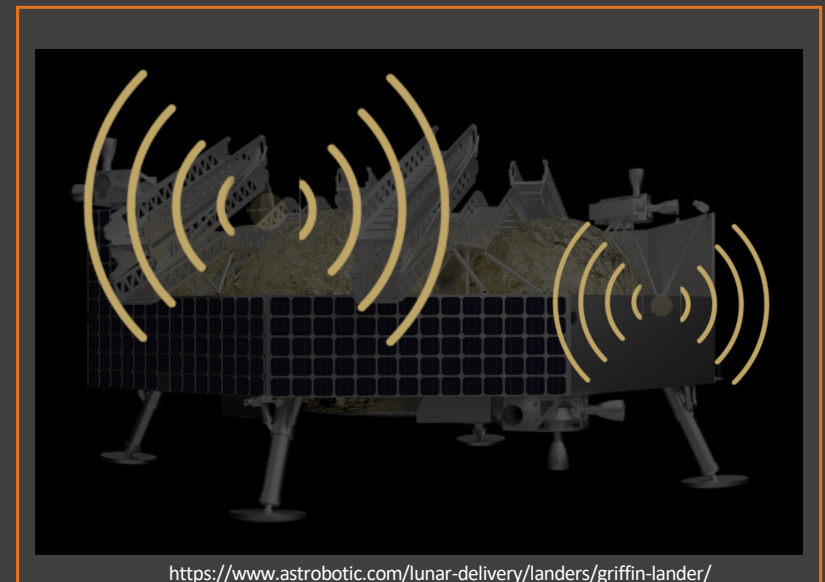
- Timeline:
 - Time per spot: 2.6 minutes
 - Time per layer: 12.6 minutes
 - Total tile time 22.3 hours
- Hardware Requirements
 - Instant power draw 1kw
 - Frequency 915 MHz
 - E field 21012.2 V/m
 - Applicator area 0.0163 m²
- Total energy required 22.3 kWh
- Scalable linear relationship (1kw-4kw)



Heating time graphed against Applicator area where the dip shows the optimal applicator area

Griffin Lander – Communication Capabilities

- According to the Astrobotic user manual, the Griffin carries a high-powered, flight-heritage transponder for Earth communications.
- Griffin has medium and high-gain antenna with a bandwidth of ~10 Kbps/kg.
- Griffin supports Serial RS-422 and SpaceWire for data communications between the lander and payloads.
- For deployed surface payloads, like rovers, Griffin provides a 2.4 GHz Wireless Local Area Network (WLAN) modem, compliant with IEEE 802.11n, for local communications. This wireless link lets surface payloads send/receive telemetry and commands via Griffin, which then relays them to Earth.
- The payload computer in Griffin is radiation-tolerant and handles both payload and lander data.



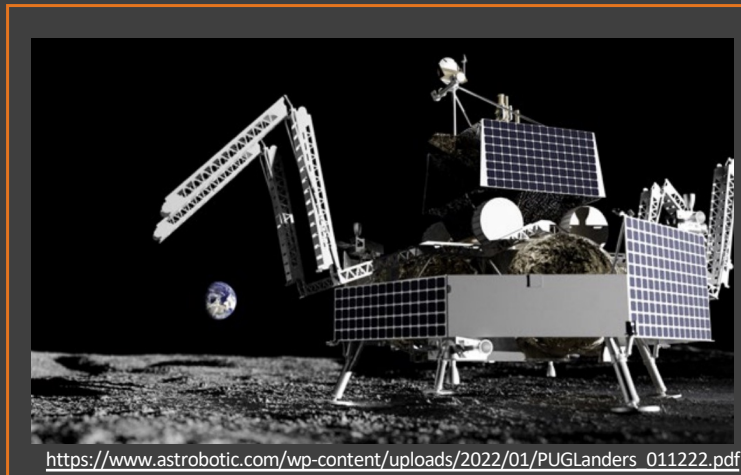
Griffin Lander – Power Capabilities

- According to the Astrobotic user manual, the Griffin Lunar Lander's power system is power positive, generating more power than it uses.
- The lander generates, stores, distributes, and manages power.
- Griffin stores energy in a space-grade lithium-ion battery. A panel of GaInP/GaAs/Ge triple junction solar cells with heritage in orbital and deep space missions generates the lander's power.
- The battery feeds into a 28 Vdc power rail from which unregulated and regulated power is distributed to all lander subsystems and payloads.
- Griffin can be configured to carry larger solar arrays and additional batteries to maximize power capabilities during transit and on the lunar surface.
- Given the C3 RFP, we are given 5 kWe.



Why do we need these capabilities?

- These capabilities, outlined in Astrobotic's user manual for the Griffin Lander, relate to the scope of our project as the Griffin Lunar Lander can actively provide our payload a power system and communications system.



Regolith Transfer System

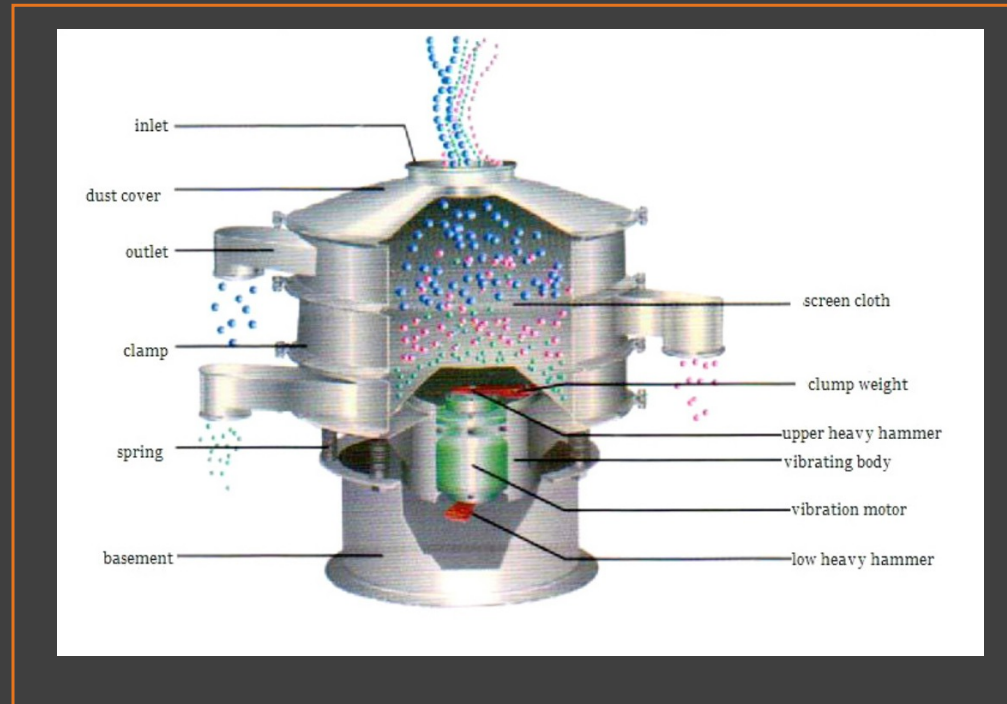
- Rover system will guide to dock and ascend a ramp
- The rover will invert its hopper and deposit regolith into larger reception hopper
- Steep graded walls ($\sim 70^\circ$) to prevent regolith accumulation
- 4 mm thick 6061-T6-Aluminum walls
- Small vibratory motor to prevent bridging
- The hopper will be capable of holding 4-5 tiles worth of regolith
 - 0.4 m^2 opening narrowing into a 0.15 m^2 inner funnel
 - Based on tile geometry the volume of the hopper is 0.03 m^3



<https://www.nbe-inc.com/processing-systems/mobile-storage-hoppers>

Regolith Conditioning – Vibratory Screener

- Top flange mates to reception hopper outlet
- Ensures optimal granularity for sintering process
- Oversized chunks sent to chute outside of tile factory
- Multiple stages
 - Stage 1: Removal of oversized rocks and clasts, durable laser cut plate
 - Stage 2: Secondary reject, too coarse for sintering, laser cut plate protects fine mesh
 - Stage 3: Final cut, fine mesh



<https://www.dahanmachine.com/news-center/Structure-And-Working-Principle-of-Circular-Vibrating-Sieve.html>

Regolith Conditioning – Vibratory Screener

Parameter		Unit	Value
Cut Size		$d_{\max}$	$600 \mu\text{m}$
Mesh Layer 1		d_{m1}	$\sim 2.0 \text{ mm}$
Mesh Layer 2		d_{m2}	$\sim 0.8 \text{ mm}$
Mesh Layer 3		d_{m3}	$500 \mu\text{m}$
Feed Rate		$\dot{m}$	0.15 kg/h
Deck Size		D	0.5 m
Frequency		f	$20 - 40 \text{ Hz}$
Motor	Power Rating	Speed	Eccentric Mass
Brushless DC	150 W	1200 – 2400 rpm	0.25 kg @ 0.3 m
Justification:			
No contact wearing, more effective in vacuum conditions	Power tied to regolith mass loads in sifter	Tied to frequency for optimal throw on earth- based sifters	Tied to required excitation force under nominal speed

Regolith Conditioning – Weigh Bin

- Top flange mates to vibratory screener
- Bin supported by 4 compression load cells
- Inlet rotary gate halts flow from screener when $m_{target} = 13.3\text{ kg}$ is reached
- Outlet slide gate releases when m_{target} is achieved, and battery is sufficiently charged for sinter cycle
- Flow spreader to distribute regolith in sinter mold



<https://www.dahanmachine.com/news-center/Structure-And-Working-Principle-of-Circular-Vibrating-Sieve.html>

Regolith Conditioning – Sintering Mold

- Controlled regolith mass dump into sintering mold
- Vibration cycle commences to ensure even distribution
- Compacting plate working in junction with vibrating plate to reach requisite bulk density of $\rho_b = 2.14 \text{ g/cm}^3$

Parameters	Unit	Value
Tile Footprint	$A = \frac{3\sqrt{3}}{2}a^2$	0.15 m
Thickness	t	10.6 cm
Vibration Frequency	f	~30 – 80 hz
Amplitude	A_{pp}	~1 mm
Compaction	σ_c	~0.4 Mpa

Regolith Conditioning – Material Selection

Subsection	Material	Choice Justification
Reception Hopper		
Shell Wall	6061-T6-Al	Lightweight, durable, Abrasion resistant, conductive
Vibratory Screener		
Casing	6061-T6-Al	Lightweight, durable, Abrasion resistant, conductive
Mesh	316L Stainless Steel	High corrosion and abrasion resistance. Function at high temperatures
Seals	Perfluoroelastomer	Excels under vacuum conditions
Weigh Bin		
Shell Wall	6061-T6-Al	Lightweight, durable, Abrasion resistant, conductive
Sintering Mold		
Outer Casing	316L Stainless Steel	High temperature resistance, microwave reflector
Mold Liner	Zirconium Ceramic	High temperature resistance, microwave-transparent

Cooling System Selection

Cooling Method	Description & Key Points	Pros/Cons	Sources
Rapid Cooling	Material is cooled quickly after sintering, often by exposure to ambient or forced air.	Fast, energy-saving, but can cause cracks or lower strength in materials.	Journal of the Mechanical Behavior of Biomedical Materials
Single-Step Cooling	Material is cooled at a constant rate from sintering temperature to room temperature.	Simple, but may not prevent thermal gradients or stress.	
Hybrid/Assisted Cooling	Combines direct and indirect cooling (e.g., susceptor-assisted, external heat sinks).	Can improve uniform of the tiles, but adds complexity.	
Cold Sintering	Uses low temperatures and pressure with a transient liquid phase for densification.	Energy efficient, but not suitable for all materials.	
Multi-Step (Staged) Cooling	Cooling is done in controlled stages, with pauses or slower rates at critical points.	Reduces thermal stress, improves quality and uniformity.	

Cooling System

- **Multi-step, Actively Managed Design:** Combines staged temperature reduction with real-time monitoring to prevent thermal shock and ensure tile quality.
- **Staged Cooling Profile:** Controlled increments (rather than rapid cooling) are essential to prevent microcracking and deformation, specifically in vacuum/low-gravity environments.
- **Active Monitoring & Feedback:** Embedded sensors provide real-time data, allowing code-based process informatics to optimize cooling schedules for specific regolith compositions.
- **Assisted Cooling Hardware:**
 - Thermal Sink Plates: High-conductivity materials help absorb heat and speed up cool-down.
 - Insulation: Ceramic fiber reduces unwanted heat loss and ensures uniform temperature gradients.

Rover components

Hopper

- Will store the regolith until it reaches the stationary tile maker
- Will tip when it reaches the stationary tile maker

Arm

- Will dig into the moons surface to collect the regolith
- Will move directly over the hopper and dump the regolith

Navigation

- Will guide the rover to areas with desirable regolith and back to the stationary tile maker
- Will tell the rover when to avoid certain areas due to obstacles

Computer systems

- Controls the motors for the arm and hopper
- Controls the driving function
- Controls the autonomy of the rover
- Supervisors
Safety / Power
of the motors

Tires

- Will be used to control the impact the regolith in the hopper will feel as the rover drives on the rough terrain

Arm Summary

Arm Requirements

- Collect 3 kg regolith per scoop
- Reach ground in front of rover and dump into hopper
- Maintain rover mass < 50 lb

Arm Architecture

- 3 DOF arm: Shoulder Pitch, Elbow Pitch, Wrist Pitch
- Link lengths:
 - Upper arm: 0.40 m, 0.7 kg
 - Forearm: 0.30 m, 0.5 kg
- End effector: 0.5 kg aluminum scoop
- Max payload at wrist: 3.5 kg (soil + bucket)

Worst-Case Posture (Static Analysis)

- Arm fully extended horizontally
- All gravitational loads acting at maximum lever arms
- Used for motor sizing (testing on Earth, not lunar gravity)

Key Lever Arms (for torque calculations)

- Payload from shoulder: 0.70 m
- Payload from elbow: 0.30 m
- Wrist offset: 0.10 m

Arm Design

Joint	Required Torque	Reason	Motor + Gearbox Class (DC)	Motor Power	Motor Mass	All Sources
Shoulder	60 N·m	Lifts entire arm and 3.5 kg scoop at 0.7 m	24 V brushed DC gearmotor with planetary gearbox 80:1–100:1	80–100 W	0.7–1.2 kg	[Mit] [Harmonic Drive] [groschopp] [AndyMark] [MABUCHI MOTOR]. [Grainger Industrial Supply]
Elbow	25–30 N·m	Lifts forearm and payload at 0.3 m	24 V brushed DC gearmotor with planetary gearbox 50:1–70:1	40–60 W	0.4–0.8 kg	
Wrist	7 N·m	Tilts bucket with 3.5 kg load at 0.1 m	12–24 V brushed DC gearmotor with planetary gearbox 20:1–30:1	10–20 W	0.2–0.4 kg	

Rover Hopper

- The Rover Hopper will be designed to hold 16 kgs of regolith to account for the rough terrain, meaning the hopper will be 35 × 25 × 12 cm.
- To the right the material chart shows the best option for the hopper is Aluminum Alloy.
- The Hopper will have walls 2.0 mm thick and floors 2.5 mm thick, making its mass is 1.36 kgs.

Material	Pros	Cons	Sources
Aluminum Alloy	• Low Density (2.66-2.8g/cm³) • Tested Lunar regolith simulants • Easy to machine • Good strength & stiffness	• Test show wear under Lunar dust stimulants	[ASM] [OnlineMetals] [Heliophysical Explorers][HELEX]
Titanium Alloy	• Durable under abrasions	• Dense (4.4- 4.8 g/cm³) • Difficult and expensive to machine	•[MatWeb] •[azom] •[sandvik]
Stainless steel	• Tough and wear-resistant • Tolerant to impacts	• Dense (7.5- 8 g/cm³) • Harder to machine than Aluminum	•[ASM] •[OBERG] •[MatWeb]

Tires

- As it can be seen from the chart the right rigid aluminum wheels with grousers are the best option.
- Based off previous rover designs it can be assumed the tires will be 0.8 kg per rigid aluminum wheel meaning the total for 4 wheels will be 3.2 kg.
- They will have a 0.27 m diameter and a width of 0.08 m ($\approx 10.5'' \times 3.1''$).

Tire Type	Pros	Cons	Sources
Rigid Metal Wheel w/ Grousers	• Simple to manufacture • Strong, durable • High traction in loose regolith • Stable under load	• More vibration/shock • Sinkage possible if grousers not optimized • Abrasion from regolith	[NASA] [Wiley]
Apollo-Style Wire Mesh Wheel	• Very lightweight • Built-in flexibility (acts like suspension) • Low sinkage • Excellent regolith performance	• Complex fabrication • Requires special materials • Mesh fatigue risk	[NASA] [Apollo Lunar Surface Journal] [NDAR]
NASA SMA / Spring Tire	• Highly compliant • Handles large deformation • Extremely durable • Excellent traction	• Very complex • Requires SMA (Nitinol) • Not low-cost	[NASA]

Navigation

- The chosen navigation is Wheel odometry and IMU Rationale because it is light, simple, heritage from MER/Mars rovers, enough accuracy for short-range excavation tasks.
- The mass based off previous systems will be less then 1 kg. This will be further explored next semester.

Navigation Method	Pros	Cons	Sources
Wheel Odometry + IMU / Dead-Reckoning	• Very lightweight, low-power, minimal hardware (just wheel encoders + IMU) • Always works (no dependency on lighting or external landmarks) • Simple to implement	• Accumulates drift over time • Wheel slip or regolith sloughing leads to big errors • No absolute referencing — position error grows unbounded without correction	[NASA]
Visual Odometry (Stereo / Camera-based)	• Can correct drift by relying on environment visual features • No moving parts or special rugged hardware (just cameras) • Good for moderate-range rover navigation over variable terrain	• Requires enough visual texture / contrast • Sensitive to lighting changes or dust covering features • Computationally heavier than odometry/IMU	[NASA] [JPL] [Wiley] [IEEE]
LiDAR or Visual-LiDAR Odometry / SLAM (3D mapping + pose estimation)	• Provides 3D terrain awareness — good for hazard detection and path planning • More robust to wheel slip or regolith interaction issues • Can build local map + correct drift repeatedly	• Heavier hardware • Requires more power, data processing, and memory • Dust/abrasion/terrain irregularities on lunar surface may degrade sensor performance	[NASA] [IEEE]

Computer Systems

Subsystem / Computer	Hardware Type	Primary Responsibilities	Key Sensors / Actuators Connected	Mass	Sources
Main Autonomy Computer	SBC (Jetson Orin / Xavier / Orin Nano)	• High-level autonomy (dig → dump) • Obstacle detection & avoidance • Mapping & path planning (ROS2) • Perception (camera/depth/LiDAR) • Localization (EKF fusion)	• Cameras (RGB/Depth) • LiDAR / Stereo • IMU • Odometry from Drive MCU	0.15–0.25 kg	[NVIDIA]
Drive Motor Controller (MCU)	Microcontroller (STM32 / Teensy / Arduino class)	• Closed-loop wheel motor control • Reads wheel encoders • Provides odometry to main computer • Executes velocity commands	• Wheel motors • Wheel encoders	0.02–0.08 kg	[Teensy] [STMicroelectronics] [Pololu] [Nav2]
Arm & Hopper Controller (MCU)	Microcontroller (STM32 / Teensy / Arduino class)	• Controls shoulder, elbow, wrist • Executes scoop cycle • Controls hopper tipping motor • Runs joint trajectories + limit switches	• Arm motors • Hopper tip motor • Joint encoders + limit switches	0.02–0.08 kg	[DYNAMIXEL] [Encoder Library]
Safety / Power Supervisory MCU	Small MCU	• Monitors battery voltage, temps • Handles E-stop • Watches “heartbeat” from main SBC • Can cut motor power if SBC fails	• Battery sensors • Power relays / MOSFETs	0.01–0.04 kg	[STMicroelectronics] [Arduino] [Adafruit] [TI]

Power Conclusions

- **Battery type:** a 24 V, 20 Ah lithium-ion deep-cycle pack — gives ~480 Wh, enough to power rover electronics, motors, and actuators for 30–60 min with margin.
- **Typical mass of this battery** is around **3.6–4.9 kg**, which fits into the mass requirements.

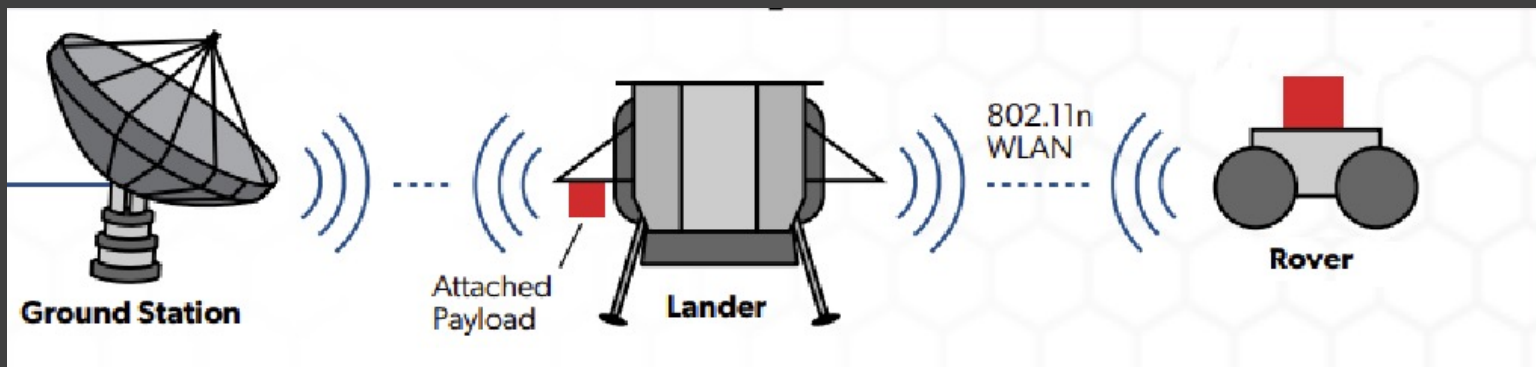


This photo is from [24v 20ah deep cycle lithium lifepo4 battery | Keheng](#)

It is being used to show the size of the battery.

Rover Communications

- High gain antennas offer direct rover to Earth communication however add mass and complexity. Instead, we will use a low gain antenna and use the Griffing Lander as a relay.
- For wireless communication, the Griffin Lander uses a 2.4 GHz WLAN modem, compliant with IEEE 802.11n. [Astrobotic]



https://www.astrobotic.com/wp-content/uploads/2024/01/Astrobotic_CubeRover-PUG_V2-2.pdf

5) Architecture Analysis

Mass Budget

Main Subsystems	Component	Mass (kg)	Growth Allowance	Growth (kg)	Total Mass (kg)
1.0 Structural & Mechanical	Rover: Chassis & Frame	4	0.15	0.6	4.6
	Rover: Wheels (4x Rigid Al)	1.8	0.1	0.18	1.98
	Rover: Excavation Arm	3	0.2	0.6	3.6
	Rover: Hopper Structure/Mech	2.8	0.15	0.42	3.22
	Tile Maker: Chassis & Frame	30	.25	7.5	37.5
	Tile Maker: Reception Hopper	6	0.25	1.5	7.5
	Tile Maker: Telescoping Ramp	6	0.25	1.5	7.5
	Tile Maker: Frame/Support	15	0.3	4.95	19.95
	Structural Subtotal				85.85
2.0 Power	Rover: Battery Pack (500Wh)	3	0.1	0.3	3.3
	Rover: Solar Panels	1	0.1	0.1	1.1
	Tile Maker: Power Harness/Cables	TBD	0.2	-	-
	Tile Maker: PDU/Regulation	TBD	0.2	-	-
	Power Subtotal				4.4
3.0 Microwave Sintering	Solid-state low-power generator	30	0.15	4.5	34.5
	Phase Array Antenna	10	0.15	1.5	11.5
	Sintering Subtotal				46

Mass Budget

Main Subsystems	Component	Mass (kg)	Growth Allowance	Growth (kg)	Total Mass (kg)
4.0 Thermal	Rover: Thermal/Dust Protection	1.5	0.25	0.38	1.88
	Tile Maker: Ceramic Fiber Insulation	TBD	0.3	-	-
	Tile Maker: Thermal Sink Plates	TBD	0.25	-	-
	Thermal Subtotal				1.88
5.0 Comm & CDH	Rover: Main Computer	0.3	0.1	0.03	0.33
	Rover: Nav Sensors/IMU	0.5	0.15	0.05	0.55
	Rover: Comms (Antenna + Transceiver)	0.3	0.15	0.05	0.35
	Tile Maker: Sensors (Temp/Mass)	TBD	0.2	-	-
	Comm/CDH Subtotal				1.23
6.0 Payload/Processing	Rover: Drive Motors (x4)	2	0.15	0.3	2.3
	Tile Maker: Vibratory Screen Assembly	12	0.3	3.6	15.6
	Tile Maker: Weigh Bin System	5	0.25	1.25	6.25
	Tile Maker: Microwave Sintering Unit	TBD	0.4	-	-
	Payload/Processing Subtotal				24.15
Sum of Totals					163.51

Power Budget – Stationary "Factory" + Rover

	Stationary Factory	
Mode	Description	Power Draw
Conditioning	- Vibratory methods - Release methods	250 W
Sintering	1 kW RF - Press - Controls	1.5 kW
Ejecting bricks	Move finished bricks off the "factory"	0.4 kW
Idle	- Day – electronics ready - Night – Survival heat + comms	50 W < 10 W

	Rover	
Mode	Description	Power Draw
Controls	- Driving - Dumping - comms	< 200 W
Digging/ Scoop	Collecting Regolith	< 150 W
Total Battery	Enough power for full day multiple trips	480 Wh

Cost Analysis

Main Subsystems	Component	Cost Estimate	Sources
1.0 Structural & Mechanical	Rover: Chassis & Frame	\$50,000	Frame estimation: https://www.endurosat.com/products/3u-cubesat-structure/
	Rover: Wheels (4x Rigid Al)	\$30,000	Wheels estimation: https://www.robotshop.com/products
	Rover: Excavation Arm	\$600,000	Excavation arm estimation: https://spacenews.com/astrobotic
	Rover: Hopper Structure/Mech	\$45,000	Hopper estimation: https://www.directindustry.com/industrial-manufacturer/hopper-dosing-dispenser-135455.html
	Tile Maker: Reception Hopper	\$50,000	Hopper estimation: https://www.directindustry.com/industrial-manufacturer/hopper-dosing-dispenser-135455.html
	Tile Maker: Telescoping Ramp	\$15,000	Ramp estimation: https://www.discounttramps.com/commercial/loading-dock
	Tile Maker: Frame/Support	\$100,000	Frame estimation: https://www.endurosat.com/products/3u-cubesat-structure/
		Total: \$890,000	
2.0 Power	Rover: Battery Pack (500Wh)	\$70,000	Lithium-Ion battery estimation: https://www.renogy.com/blogs/buyers-guide
	Tile Maker: Power Harness/Cables	\$7,500	Cable estimation: https://www.weicowire.com/products
	Tile Maker: PDU/Regulation	\$50,000	Power distribution unit estimation: http://www.marway.com/applications
		Total: \$127,500	

Cost Analysis

Main Subsystems	Component	Cost Estimate	Sources
3.0 Thermal	Rover: Thermal/Dust Protection	\$200,000	Protection estimate: https://ntrs.nasa.gov/api/citations/20220018746/downloads/TP-20220018746.pdf
	Tile Maker: Ceramic Fiber Insulation	\$40,000	Ceramic fiber estimation: https://ntrs.nasa.gov/api/citations/19990047691/downloads/19990047691.pdf
	Tile Maker: Thermal Sink Plates	\$40,000	Thermal sink plate estimations: https://www.nasa.gov/smallsat-institute/sst-soa/thermal-control/
		Total: \$280,000	
4.0 Comm & CDH	Rover: Main Computer	\$350,000	Main computer estimation: https://www.spaceflightnow.com/mars/msl/120810computer/
	Rover: Nav Sensors/IMU	\$75,000	Sensor estimation: https://www.cubesatshop.com/product-category/attitude-sensors/
	Rover: Comms (Antenna + Transceiver)	\$60,000	WGM160P Wi-Fi® Module Data: https://www.silabs.com/documents/public/data
	Tile Maker: Sensors (Temp/Mass)	\$15,000	Sensor estimation: https://www.cubesatshop.com/product-category/attitude-sensors/
		Total: \$500,000	
5.0 Payload/Processing	Rover: Drive Motors (x4)	\$60,000	40–60 W planetary gearmotors: https://www.groschopp.com/motors
	Tile Maker: Vibratory Screen Assembly	\$30,000	Vibratory screen estimation: https://www.shake-it.com/product_category
	Tile Maker: Weigh Bin System	\$35,000	Weigh bin estimation: https://www.directindustry.com/industrial-manufacturer/hopper-dosing-dispenser-135455.html
	Tile Maker: Microwave Sintering Unit	\$100,000	Sintering unit estimation: https://hulc.nianet.org/wp-content
		Total: \$225,000	
		Complete Total: \$2,022,000	

Risk Assessment

Consequence of Failure		Likelihood of Failure	
Catastrophic (5)	Mission fails to produce landing pad, and any technical demonstrations.	Very High (5)	81% - 100%
Critical (4)	Mission fails to demonstrate capability of landing pad, few results collected	High (4)	61% - 80%
Moderate (3)	Mission experiences reduced performance in one of mission critical systems	Medium (3)	41% - 60%
Minor (2)	Mission experiences reduced performance affecting one of the noncritical systems	Low (2)	21% - 40%
Negligible (1)	Mission requires slight changes after launch	Very Low (1)	0% - 20%

Risk Classification	Category
M	Mass
G	General
E	Environmental
P	Power
Sc	Schedule
St	Structural

Mission Risk Analysis

Risk ID	Risk Statement	Consequence	Likelihood	Mitigation	Consequence After Mitigation	Likelihood After Mitigation
M1	Surpassing available mass	5	3	Use conservative mass estimates	5	1
Sc1	Schedule overrun	2	4	Use conservative time estimates and budget accordingly	1	4
G1	Solar panels dust accumulation	3	3	Controlled dumping of regolith and dust shield	3	1
P1	Unexpectedly low microwave efficiency	2	4	Extensive testing beforehand	2	1
St1	Rover is unable to traverse terrain and gets stuck	5	2	Have chosen a flat landing/collecting site	5	1
G2	Change in regolith composition in the landing site	4	4	Choose a location with stable and predictable regolith	4	2
E1	Radiation overheats microwave	4	3	Implementing a cooling system	2	3

Consequence of Failure	5	←	M1	→			
	4	←	St1	←		G2	
	3	←	G1	←	E1		
	2	←		P1	→	Sc1	
	1						
		1	2	3	4	5	
		Likelihood of Failure					

Risk Matrix

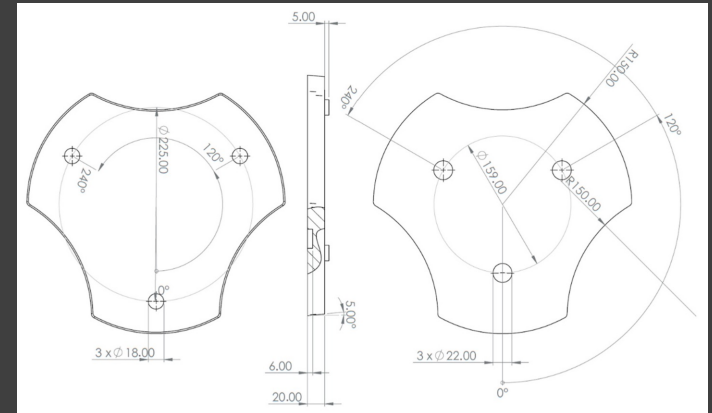
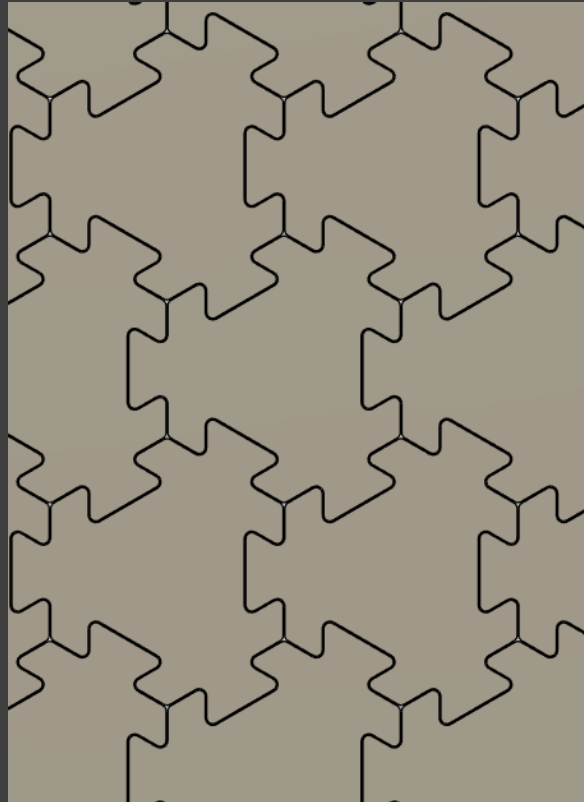
Looking forward to the future

For the future:

- Our current goal is a proof of concept, saying that tiles can be made on the Moon. In the future we want a full tile factory, building a fully usable landing pad.
- We will be looking into a more autonomous rover system. Finding the feasibility of adding in the ability of the rover to move tiles away from the factory for testing.
- We want to find a full proof way of testing the tile(s) on the Moon, in addition of shipping the tiles back to Earth.
- Perform Finite Element Analysis on different tile geometries to find an optimal solution.
- We want to add an interlinking geometry system for the tiles to connect with each other completely.
- We want to be able to terraform a suitable landing pad area.

Interlocking Tile Designs

- We designed interlocking hexagonal tiles with a dovetail.
- Dovetails allow for transfer of lateral forces between tiles and interlock to prevent ejection under plumes during landing and takeoff.
- Currently we are unable to produce these designs due to radial shrinkage in our manufacturing process.



www.mdpi.com/2226-4310/12/9/781

- We utilized dimensions based upon above tile to withstand thermal and pressure requirements.
- The tile above was not used as it relies upon 3-dimensional joinery which our manufacturing process doesn't accommodate due to vertical non-homogeneity.

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NASA Technical Reports Server (NTRS)
Slide 66
NASA Technical Reports Server (NTRS)

Slide 67

Pololu - Brushed DC Motor Controllers
<http://www.pololu.com>

Arduino Official Store | Boards Shields Kits Accessor...

DYNAMIXEL

EncoderLibrary for Measuring Quadrature Encoded Position or Rotation Signals

Questions?

Aerospace and Ocean Engineering Dept
Virginia Polytechnic Institute and State University
AOE 4165: Space Vehicle Design
Fall 2025 Semester



Extra Slides

Wireless Node

- We plan to use a flight-qualified wireless charging node developed by Astrobotic and WiBotic
- The node uses magnetic-resonance wireless power transfer
- It can deliver up to ~125 W
- Because it's wireless, it avoids dust-plug problems, making repeated charging and long-term operations on the Moon feasible

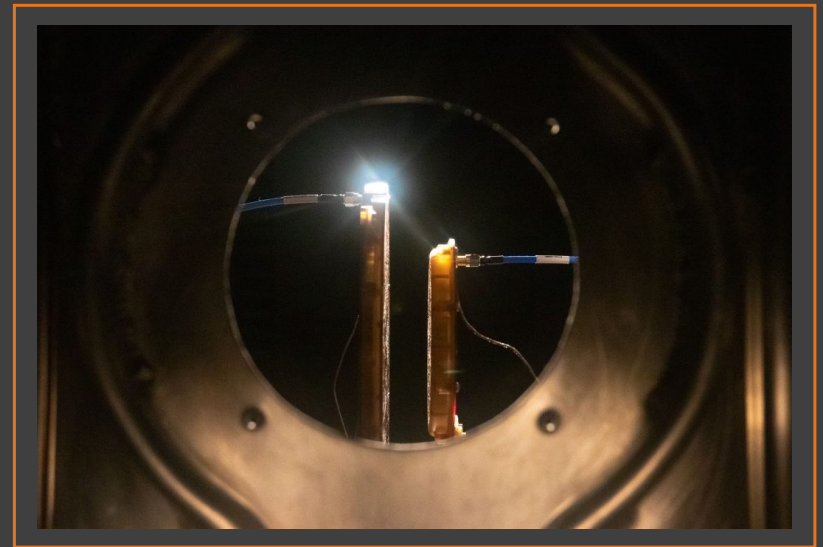
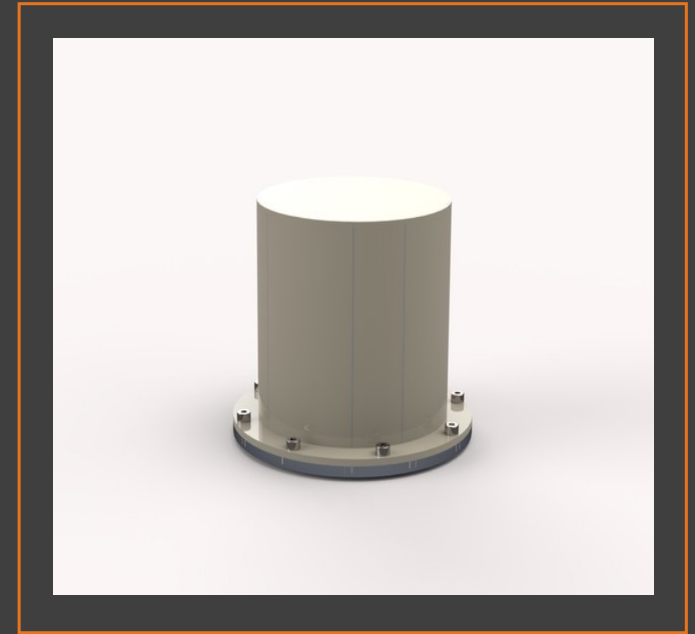


Image: Wireless charging nodes inside TVAC chamber.

<https://www.astrobotic.com/astrobotic-lunar-wireless-charger-system-qualified-for-flight/>

Rover Communications

- Silicon Labs WGM160P Module [Silicon Labs]
 - Compatible with IEEE 802.11n
 - Maximum power requirement of 0.5 Watts
- 2.4 GHz Pagoda Antenna [Satsearch]
 - Will use 2 for redundancy and full coverage [ESA]
 - 0.6 - 1.3 dBi gain ensures constant reception and transmission
- Multi-layer Insulation [NASA]
 - Composed of Kapton and aerogel
 - Lightweight material



<https://satsearch.co/products/eosol-group-pagoda-antenna-for-lunar-communications>

Hopper Motor

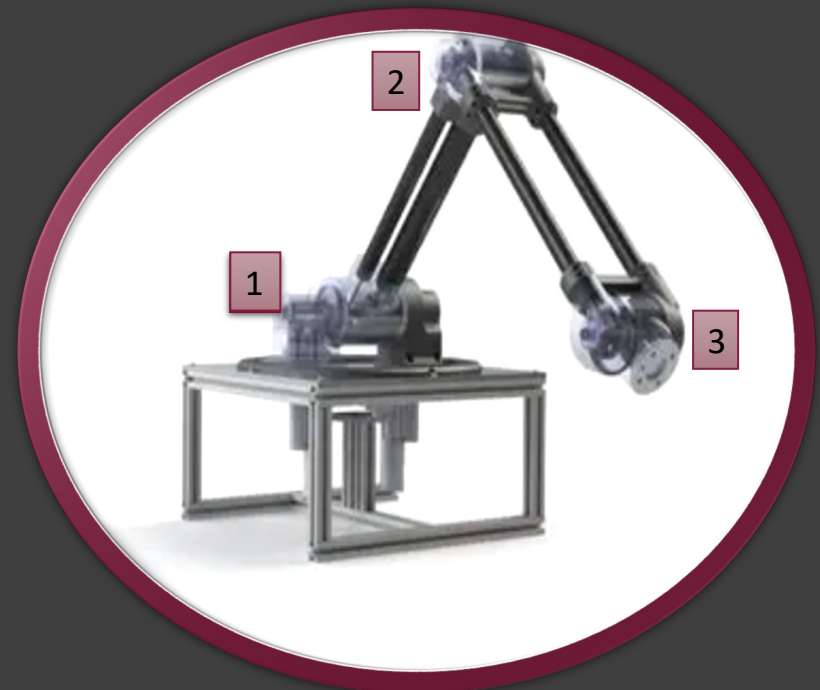
Item	Value
Lever arm (pivot → CoM)	~0.175 m (half of 35 cm hopper length)
Required torque (no SF)	~23–28 N·m (for 12–15 kg regolith + hopper)
Design torque (with SF ≈ 2)	~50–60 N·m at hinge
Motor type	40–80 W DC gearmotor + mount
Gear ratio	~50:1–80:1
Motor power (approx)	40–80 W
Motor mass (approx)	.8 kg
Sources	Spong, Hutchinson, Vidyasagar — Robot Modeling and Control DC Planetary Gear Motors – RobotShop Pololu Robotics and Electronics MABUCHI MOTOR CO., LTD. Stepper Motor, Servo Motor, BLDC Motor, Gearbox – StepperOnline Educational Robotic Parts - AndyMark, Inc DC Planetary Gearmotors for Efficiency and High Torque – Groschopp NASA Technical Reports Server (NTRS) Introduction to Robotics Mechanical Engineering MIT OpenCourseWare A New Remote Sensing Filter Radiometer - NASA Technical Reports Server (NTRS)

Power

Subsystem / Component	Peak Power (W)	Notes
Main Autonomy Computer (Jetson Orin Nano 8GB)	15 W	NVIDIA lists Orin Nano modules operating between 7–15 W; 15 W is peak. NVIDIA Jetson Orin Nano 8 GB: Detailed Specifications and Benchmark Ratings - CpuTronic
Drive Motor Controller MCU (STM32/Nucleo-64)	1 W	Typical runtime draw in tens of mA; 1 W is a conservative design top. STM32 Nucleo boards - STMicroelectronics
Arm & Hopper Controller MCU (Teensy 4.1 / STM32)	1 W	Teensy 4.1 draws ~100 mA at 3.3 V (~0.33 W); 1 W covers all peripherals and overhead. Teensy® 4.1
Safety/Power MCU	0.5 W	Small microcontrollers typically <0.2 W; 0.5 W gives margin. STMicroelectronics: Our technology starts with you
Radio / Communications Module (XBee-PRO 900HP)	1 W	Datasheet: TX current ≈ 290 mA @ 3.3 V → ~0.96 W. Rounded to 1 W. XBee®-PRO 900HP/XSC RF Modules
Sensor Suite (IMU + camera(s) + depth/LiDAR optional)	10 W	Typical small robot sensor stack; depth camera adds ~2–4 W, IMU negligible, cameras 1–3 W each. Buy a Raspberry Pi Camera Module 2 – Raspberry Pi Intel® RealSense™ LiDAR Camera L515 Datasheet
Drive Motors (4 DC gearmotors)	160 W	Assuming ~40 W peak per wheel under load on regolith. DC Planetary Gearmotors for Efficiency and High Torque - Groschopp
Arm Motors (shoulder, elbow, wrist)	~100 W	Shoulder and elbow in 40–80 W class; wrist small. Not all at max simultaneously. Gear Motor Selection Guide (How to choose high torque Gear Motor) - Best Gear Motor
Hopper Tip Motor	50 W	Matches torque requirement (~50–60 Nm) with a 40–80 W DC gearmotor. Electric Motors Industrial Motors AutomationDirect
TOTAL Peak Power	~339 W	everything running at max power at once

Arm Design

Joint	Degrees of Freedom	Purpose
Joint 1	Shoulder Pitch	Raises/lowers entire arm
Joint 2	Elbow Pitch	Extends/retracts reach
Joint 3	Wrist Tilt	Controls scoop angle for digging



This photo is from [Lunar Rover Arm Brings Increased Functionality to space missions.](#)
It is being to show the design of the rover arm.

Subsystem / Part	Assumptions / Notes	Est. Mass (kg)
Chassis / Frame	Aluminum frame, brackets, mounting plates ASM Metals Handbook — Aluminum 6061 Properties. NASA KSC Lunabotics Design Case Studies. 80/20 Aluminum T-slot Building Systems 80/20 Aluminum Extrusions	3.0
Wheels + Drive Motors (4x)	Rigid aluminum wheels with grousers + 4 DC gearmotors + mounts NASA Regolith Mobility Challenge whitepapers. DC Planetary Gear Motors - RobotShop	4.0
Arm Structure (links, brackets)	Two aluminum links (0.4 m + 0.3 m), wrist housing, mounting hardware OnlineMetals.com ® Online Metal Supplier Buy Metal and Raw Materials from JPL Robotic Arm Prototyping Reports.	1.2
Arm Motors (shoulder, elbow, wrist)	2 medium gearmotors (~0.7 kg each) + 1 small (~0.3 kg) DC Planetary Gearmotors for Efficiency and High Torque - Groschopp	1.7
Hopper Structure	calculated hopper: 35x25x12 cm, 2.0/2.5 mm aluminum → already 1.36 kg A Periglacial Analog for Landforms in Gale Crater, Mars - NASA Technical Reports Server (NTRS) ASM Handbook: Aluminum Alloys.	1.36
Hopper Tip Motor / Actuator	One 40–80 W DC gearmotor + mount RobotShop Robot Store Humanoids Robots Robot Parts & Kits	0.8
Main Battery Pack (24 V, ~20 Ah)	24 V, 20 Ah LiFePO ₄ /NMC pack, ~3.6–4.9 kg typical (take middle) Aegis 24V 20Ah Lithium Ion Battery Pack NMC 24V Lithium Battery Aegis Battery	4.0
Main Computer (Jetson / Pi + carrier)	Jetson Orin Nano dev kit (=0.17–0.2 kg) or Pi + carrier NVIDIA Developer	0.2
MCUs (Drive, Arm/Hopper, Safety)	3 small STM32/Teensy boards, wiring, mounts STMicroelectronics: Our technology starts with you Teensy® 4.1	0.15
Sensors (IMU, cameras, LiDAR)	1 IMU, 1–2 small cameras, maybe a lightweight LiDAR/depth cam Intel® RealSense™ Technology	0.6
Thermal / Dust Protection	Shields, light insulation, motor covers NASA Thermal Systems for Lunar Surface Robotics	1.0
Wiring, Fasteners, Misc. Hardware	Cable harnesses, connectors, bolts, spacers, labels JPL Pathfinder Wiring Mass Summary. McMaster-Carr bolt weight charts.	1.2
Communication Systems	Antenna, modem, mounting hardware, connectors and wiring, Insulation XBee®-PRO 900HP/XS-C RF Modules	0.2
Weight total		19.41

